PROJECT MANAGEMENT CONSULTANCY | GMDC COAL BLOCKS, ODISHA

Coal Mining PMC

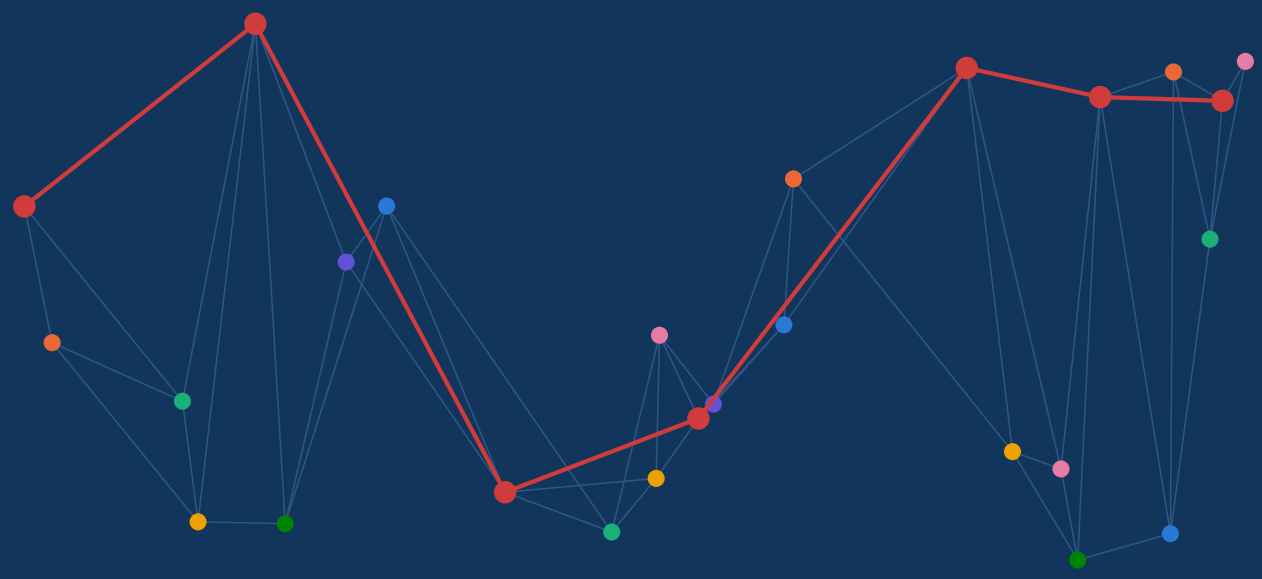
From block auction to mine operationalisation in Odisha

End-to-end process map (CBA Act land route) • ~200 granular activities
PERT three-point estimates • CPM forward/backward pass • Monte Carlo risk
Critical-activity register • Baitarani West status & forecast

CASE STUDY

Baitarani West Coal Block - Talcher Coalfield, Angul, Odisha

15 MTPA opencast • 1,196 ha • 468 Mt extractable • G12 • 37-year life
Successful bidder: Gujarat Mineral Development Corporation Ltd (GMDC)
CMDPA 29-Mar-2023 • Vesting order 08-Jun-2023 • Final Offer 33.50% revenue share



Status date: 14 October 2026 • Prepared with public sources (MoC, PIB, PARIVESH, CCO, Railways) and project records

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What this report is. A practitioner's playbook for taking a commercial coal block in Odisha from the auction notice to the first tonne of coal, when land is acquired by a Government company under the **Coal Bearing Areas (Acquisition & Development) Act, 1957**. Every statutory and practical step is listed (about 200 granular activities), organised into nine process flows, then rolled up into a 62-activity network analysed with **PERT** (three-point time estimates), **CPM** (critical path and float) and a 20,000-run **Monte Carlo** simulation.

How to read it. Colours identify the seven work phases throughout (legend below). A red outline or bar always means **critical** (zero float); a navy diamond marks a CMDPA milestone (MS-2 to MS-5) or another gate. Durations are in months unless stated. Section 10 applies the method to Baitarani West with status as on 14-Oct-2026.

Sources. Project records in the repository (CMDPA, final EIA/EMP, MIS of 30-Sep-2026, 2023 tracker, bidding deck, Acts and Rules) and web research on Ministry of Coal, PIB, PARIVESH, Coal Controller, Railway Board, CGWA and Odisha Government sites (Annex B). The 2023 tracker was used for its activity lists only; its dates and several procedures are superseded (Section 12).

Auction & vesting · Mobilisation & surveys · Mining plan · Environment clearance · Forest clearance · Land acquisition (CBA) & R&R · Consents, infrastructure & mine opening · Critical path · Milestone / CMDPA gate

The nine processes on one page

Process	Legal basis	Sanctioning authority	Typical span*	CMDPA link	Baitarani West (Oct-26)
1 Auction & vesting	CM(SP) Act 2015; MMDR Act	Nominated Authority (MoC)	6-9 months	Zero date	Done - CMDPA 29-Mar-23; vesting 08-Jun-23
2 Mobilisation & surveys	MP Guidelines 2025 (DGPS)	CMPDI / ORSAC; Tehsildar	3-6 months	Feeds all	Done - DGPS certified 12-Jan-24
3 Mining plan & closure plan	MCR 2016/2020; MP Guidelines 2025	CCO / MoC	4-8 months	MS-2	Done - OMC plan transferred 13-Oct-23
4 Environment clearance	EP Act 1986; EIA Notification 2006	MoEFCC (EAC Coal)	14-24 months	MS-3	Done - EC 03-Dec-25 (~12 months late)
5 Forest clearance (+FRA, CA, WLCP)	VSSR 2023; FRA 2006; WPA 1972	MoEFCC; Odisha Forest Dept	24-40 months	Feeds MS-5	Stage-I 02-Dec-25; Stage-II at MoEFCC; FRA pending
6 Land acquisition (CBA)	CBA Act 1957	MoC; Collector as competent authority	12-26 months	MS-4	s.4/7/9/11 notified 13-Oct-23; payments ongoing
7 Resettlement & rehabilitation	RFCTLARR Sch. I-III (S.O. 2368(E)); ORRP 2006	Collector / RPDAC	26-45 months	MS-4	RPDAC held 07-Oct-26; colony & shifting pending
8 Consents & infrastructure	Water/Air Acts; CGWA; GCT Circular	OSPCB, WRD, CGWA, ECoR, PESO	6-30 months	Feeds MS-5	CTE 29-Apr-26; CTO & nala NOC pending; MDO signed 17-Sep-26
9 Mine opening & first coal	Mines Act s.16; CMR 2017; CCR 2004 (amended)	GMDC Board; DGMS; CCO (intimation)	1-6 months	MS-5	CMDPA date 08-Mar-27; P50 forecast in Section 10

** Elapsed span of the process for a fresh block, from the CPM model (Section 7); processes overlap heavily, so spans are not additive.*

AT A GLANCE

1 Executive summary

62 months
Expected duration, tender launch to first coal, for a fresh Odisha block (CPM on PERT means)

50.7 months
Expected vesting-to-mine-opening (MS-5) time vs 51 months allowed by the CMDPA

53%
Probability of opening within the 51-month window (Monte Carlo, 20,000 runs)

97%
Criticality index of the land & R&R chain - it drives the finish in almost every run

Key messages

- **The path from bid to coal has nine interlocking processes and ~200 steps**: auction & vesting, mobilisation & surveys, mining plan, environment clearance, forest clearance (with FRA, CA land, tree enumeration and wildlife plan), CBA land acquisition, R&R, other consents, and mine opening. Only four milestones are policed by the CMDPA (MS-2 mining plan, MS-3 EC, MS-4 land & R&R, MS-5 mine opening), but every other step feeds them.
- **The critical path runs through land, not clearances**: A1 → A2 → A3 → A4 → A5 → A6 → B2 → B4 → B5 → L1 → L2 → L3 → L7 → L9 → L12 → L13 → K12 → K13 → K15. Under the CBA route the slowest items are SES/genealogy/YADAST, RPDAC approval of the R&R plan, and building the R&R colony and shifting families before possession (MS-4).
- **Forest clearance is the near-critical path** (total float ≈ 10 months): CA land suitability, the multi-tier State processing, Stage-I compliance and the FRA certificate. It becomes critical as soon as land is fast-tracked (scenarios S3-S5).
- **EC, mining plan, MDO tendering and most consents carry large float** - they should still be started early, because their real risk is rework (e.g., EAC returning a proposal) rather than duration.
- **Fast-tracking works**: starting the SES at s.4(1), overlapping R&R planning with the SES and phasing possession (box cut on a no-habitation patch) lift the probability of meeting the 51-month window from 53% to 100% (S4).
- **Baitarani West (status 14-Oct-2026)**: mining plan, EC (03-Dec-25), FC Stage-I (02-Dec-25), CTE (29-Apr-26) and the MDO contract (17-Sep-26) are done; FC Stage-II, FRA certificate, nala NOC, CTO, compensation for the 5-year area and R&R shifting remain. Our P50 forecast for mine opening (MS-5) is **03-Apr-27** with phased possession (P = 12% of meeting the CMDPA date 08-Mar-27) but **29-Aug-27** if R&R shifting must precede opening (P = 55% of opening inside the 51-month window, 08-Sep-27).
- **Money is attached to the dates**: up to ₹386.8 Cr of performance security (MS-3 10% + MS-4 25%) can be appropriated for BW's delays but becomes refundable if MS-5 is reached within 51 months of vesting; every tonne produced before 08-Sep-2029 attracts only half the 33.5% revenue share.

Bottom line

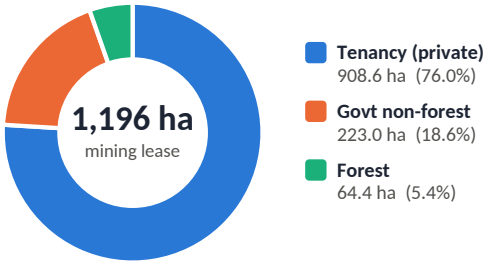
Top five actions for Baitarani West (next 90 days): (1) close FC Stage-II at MoEFCC and the FRA certificate at DLC Angul in parallel; (2) keep CTO 'ready except FC-II' so it issues within days; (3) finish the 199-ha case files and disbursement camps and take possession of the box-cut patch; (4) notify the R&R scheme after the 07-Oct-26 RPDAC and lock colony and gochar land; (5) file an extension representation to the Nominated Authority for delays not attributable to GMDC, and plan MS-5 well before 08-Sep-2027.

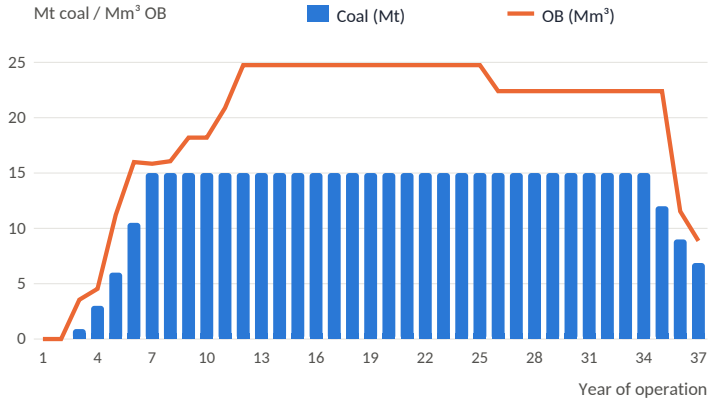
CASE STUDY

2 Baitarani West at a glance

Parameter	Value	Parameter	Value
Location	Chhendipada tehsil, Angul district, Odisha; north-central Talcher coalfield	Villages	Chhendipada Jangal, Handigora, Machhakuta, Porapara, Tentuloi Korasahi and others
Mine type / capacity	Opencast; 15 MTPA (surface miner 93%, drill-blast 7%)	Lease area	1,196 ha (tenancy 908.65; Govt 222.98; forest 64.37)
Reserves	Gross GR 662.9 Mt; net GR 602.1 Mt; extractable 468.3 Mt	Grade / strip ratio	G12 (~3,946 kcal/kg); 1.53 m^3/t
Depth / seams	160-245 m; seam thickness 0.08-33.94 m	Life	37 years; peak 15 MTPA from year 7
Dumps	External 215.8 ha (140 Mm^3, 90 m); internal 623.7 ha (573 Mm^3)	Evacuation	12 MTPA conveyor + silo + siding to ECoR (Jarpara); 3 MTPA road
Project cost	₹3,855 Cr (R&R ₹727.94 Cr; EMP capex ₹100.7 Cr)	Cost of production	₹337.11/t (EIA)
Manpower	1,752 direct (269 departmental, 1,483 contractual); ~5,000 indirect	Affected families	3,402 original families in SES (Jan-25)
Auction	16th tranche CM(SP) Act; Final Offer 33.50%	CMDPA / vesting	29-Mar-2023 / 08-Jun-2023 (NA-104/4/2023-NA)
Upfront amount	₹500 Cr in 4 × ₹125 Cr instalments	Performance security	₹1,105.19 Cr; fixed amount ₹49.40 Cr
History	Cancelled 2014 (SC); OMC allotment 2016; OMC mine plan approved 10-Nov-21	Land route	CBA Act: s.4(1), 7(1), 9(1), 11(1) in GMDC's favour 13-Oct-23

Sources: CMDPA (29-Mar-2023); Final EIA/EMP (GRC India) Tables 2.3, 2.12, 10.2; MIS 30-Sep-2026.





Left: land composition of the lease (EIA Table 2.3). Right: approved production and overburden schedule - first coal in operating year 3, 6 MTPA in year 5 (rail needed), 15 MTPA from year 7.

Money tied to milestones

Payment / security	Trigger (CMDPA)	Amount	BW status
Upfront 1st (25%) to NA	≤40 business days of CMDPA	₹125 Cr	Paid 02-May-23 (due 08-May-23)
Upfront 2nd (25%) to State	≤15 business days of s.11(1) order	₹125 Cr	Paid 04-Jan-24 (due 28-Oct-23; State-side delay)
Upfront 3rd (25%) to State	≤15 business days of mine-opening permission	₹125 Cr	Due ~Mar-27 (CMDPA basis)
Upfront 4th (25%) to State	≤15 business days of first coal	₹125 Cr	To be scheduled
Performance security	65% × (1-yr royalty + 1-yr revenue share) at PRC	₹1,105.19 Cr	With NA; moves to State at mine opening
PS appropriation risk	MS-2 10% / MS-3 10% / MS-4 25% / MS-5 25%	₹110.5 / 110.5 / 276.3 / 276.3 Cr	MS-3 late (EC 03-Dec-25); MS-4 overdue since 08-Sep-26
Monthly revenue share	33.5% × royalty quantity × max(notional, actual price)	-	Upfront adjusted ≤50% of monthly payment

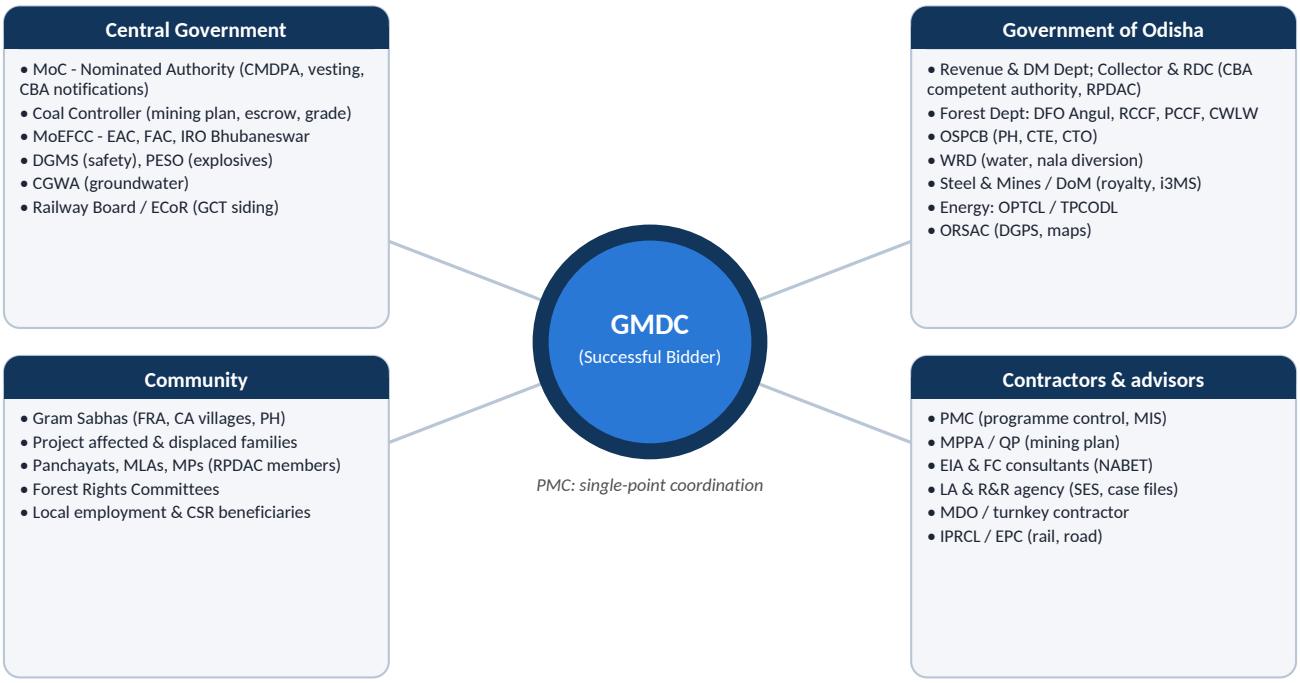
Note 6 of CMDPA Schedule D: if MS-5 is achieved within 51 months of vesting (BW: 08-Sep-2027), performance security appropriated for earlier milestone delays may be refunded.

WHO DECIDES WHAT

3 Regulatory and institutional framework

Law / instrument	What it governs for the project	Authority	Stage
Coal Mines (Special Provisions) Act, 2015 & Rules	Auction, vesting/allotment orders, Nominated Authority	MoC (NA)	Auction & vesting
Mines & Minerals (Dev. & Regulation) Act, 1957; MCR 2016/2020	Auction of coal mines, mining plan, royalty (14% ad valorem), DMF, NMET	MoC / State	Throughout
Coal Bearing Areas (Acquisition & Development) Act, 1957	Acquisition of land/rights (s.4, 7, 8, 9, 10, 11, 12, 13, 14)	MoC / Competent Authority	Land acquisition
RFCTLARR Act 2013 + Removal of Difficulties Order 2015 (S.O. 2368(E))	Compensation (Sch. I), R&R; (Sch. II), amenities (Sch. III) for CBA acquisitions	MoC / State	Land & R&R;
Odisha R&R; Policy 2006; ORFCTLARR Rules 2016	RPDAC, entitlements, procedures in Odisha	Revenue & DM Dept / Collector	Land & R&R;
Environment (Protection) Act, 1986; EIA Notification 2006	ToR, baseline, public hearing, EC (Category A, item 1(a))	MoEFCC / EAC	EC
Van (Sanrakshan Evam Samvardhan) Adhiniyam 1980 & Rules 2023	Forest diversion, CA, NPV, Stage-I/II	MoEFCC / State Forest Dept	FC
Scheduled Tribes & OTFD (Recognition of Forest Rights) Act, 2006	Settlement of forest rights; Gram Sabha consent; FRA certificate	Collector / DLC	FC
PESA Act, 1996	Gram Sabha consultation in Fifth Schedule areas (verify status of villages)	Panchayati Raj Dept	Land / FC
Wildlife (Protection) Act, 1972	Wildlife plan, elephant corridors; NBWL if within 10 km of PA/ESZ	CWLW / NBWL	FC / EC
Water Act 1974; Air Act 1981; HOWM Rules 2016	CTE, CTO, hazardous-waste authorisation	OSPCB	Consents
CGWA Guidelines 2020 (EP Act s.3)	Groundwater abstraction & dewatering NOC	CGWA (NOCAP)	Consents
Mines Act 1952; Coal Mines Regulations 2017	Notice of opening, manager appointment, safety	DGMS	Mine opening
Colliery Control Rules 2004 (amended 2025 & 2026)	Mine/seam opening by Board; intimation to CCO; grade declaration	MoC / CCO	Mine opening
Explosives Act 1884 & Rules 2008; Petroleum Rules 2002	Magazine and HSD storage licences	PESO	Mine opening
CLRA 1970; CMPF Act 1948; BOCW Act 1996	Contract labour, provident fund for coal-mine workers	RLC (C) / CMPFO	Mine opening
Indian Railways GCT Master Circular 2026	Private siding / cargo terminal approvals	ECoR / Railway Board	Evacuation
Companies Act 2013 (s.135)	Board approvals, CSR	GMDC Board	Throughout

Stakeholder map



GMDC sits at the centre; the PMC's job is to keep ~25 counterparties moving on one integrated schedule.

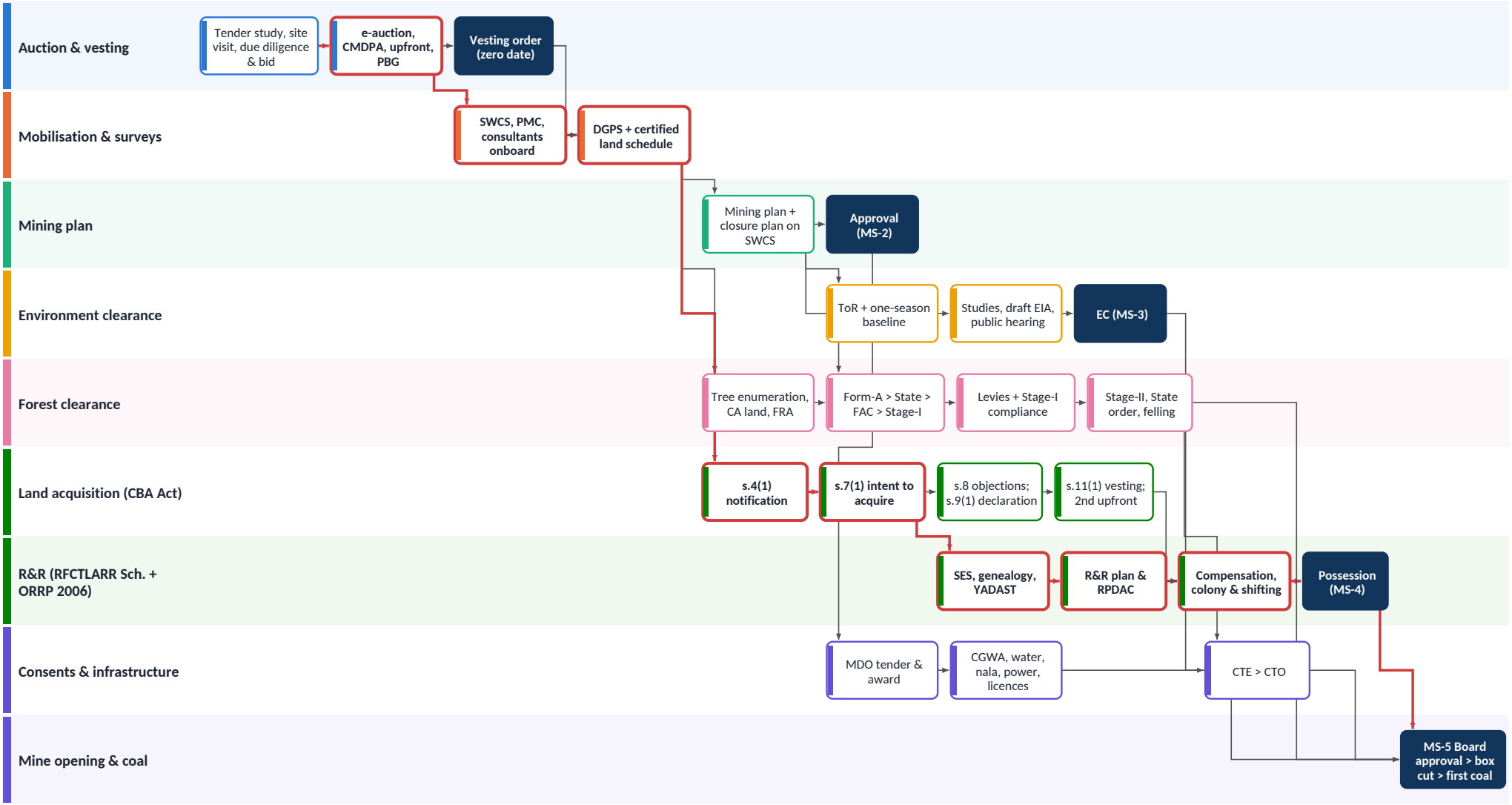
Responsibility matrix (RACI)

Process	GMDC	PMC	Consultants	MDO	MoC / NA / CCO	MoEFCC	Collector / Revenue	Forest Dept	OSPCB / WRD / others	Gram Sabha / PAFs
Auction & vesting	A R	C	C		S					
Mobilisation & surveys	A	R	R		C		S	C	C	I
Mining plan (MS-2)	A	C	R	I	S					
Environment clearance (MS-3)	A	C	R	I	I	S	C	C	R	C
Forest clearance	A	C	R		I	S	R	S		C
CBA land acquisition	A	C	R		S		R			C
R&R; (MS-4)	A	C	R	I	I		S			C
Consents & infrastructure	A	C	R	R	I		C		S	I
Mine opening (MS-5) & coal	A S	R	C	R	I		I		S	I

A = accountable, R = responsible (does the work), S = sanctions / approves, C = consulted, I = informed. Government agencies 'sanction'; GMDC stays accountable for every row - including the Board's own sanction of mine opening under the amended Colliery Control Rules.

FLOW OF ALL PROCESSES

4 End-to-end process map



Logic of the whole programme. Red = critical chain from the CPM; navy boxes = milestones. Horizontal position is indicative of sequence, not to scale. Section 5 expands every box into its detailed flow and step list.

CMDPA efficiency parameters - the four policed milestones

MS	Main activity monitored	Time allowed (fully explored)	PS appropriated if late	BW due date	BW actual / forecast
MS-1	Geological Report (partially explored only)	Not applicable	-	-	-
MS-2	Approval of Mining Plan	6 months from allocation	10%	08-Dec-23	Met - OMC plan deemed transferred
MS-3	Environment Clearance	18 months after MS-2	10%	08-Dec-24	03-Dec-25 (~12 months late)
MS-4	Land acquisition, possession & R&R for rated capacity	21 months after MS-3	25%	08-Sep-26	Overdue; P50 13-Aug-27 incl. shifting
MS-5	Mine opening permission	6 months after MS-4	25%	08-Mar-27	P50 03-Apr-27 (phased)
Total	All milestones	51 months (66 if partially explored)	Refund of earlier PS if MS-5 in 51 months	08-Sep-27	-

Non-monitored Schedule D activities (no penalty, but tracked for early development): mining lease application, FC and EC applications, FC Stage-I/II, wildlife clearance, PESA approval, nala/river diversion, diversion of power line/rail/road, permission to draw water and power, CTE/CTO, grant of ML or s.11 CBA order, DGMS intimation, explosives and petroleum licences, Factories Act permission, RF communication licence, labour permissions and the escrow account. Schedule D note 3 expressly allows clearances to be obtained concurrently.

Reading the clock. The MIS (and this report) treats each milestone's due date as fixed from the vesting date (MS-2 at vesting since the plan was transferred; MS-3 = vesting + 18 months, and so on). Schedule D says 'from completion of previous MS', which a late MS-3 could arguably reset; this conservative reading should be confirmed with the Nominated Authority before any extension request.

Statutory and contractual clocks - quick reference

Area	Step	Time limit	Source
Auction & CMDPA	Bid security validity	210 days from bid due date + 60-day claim period	Tender Document
Auction & CMDPA	1st upfront instalment (25%)	Within 40 business days of executing the CMDPA	CMDPA cl. 5.2.1
Auction & CMDPA	2nd / 3rd / 4th instalments	Within 15 business days of s.11(1) order (or ML) / mine opening / first coal	CMDPA cl. 5.2.1
Auction & CMDPA	PBG substitution on revision	Within 30 days of the trigger; annual revision by 30 April after opening	CMDPA cl. 6.1.5
Auction & CMDPA	Efficiency parameters	MS-2 6 m; MS-3 +18 m; MS-4 +21 m; MS-5 +6 m; total 51 m (66 m if partially explored)	CMDPA Sch. D
Auction & CMDPA	Scheduled Date of Production	75 months from vesting (end of 50% early-production rebate)	CMDPA cl. 9.2.1
Auction & CMDPA	Minimum production	≥65% of plan each FY; ≥75% over each 3-FY block; shortfall paid in 20 days	CMDPA cl. 16.2
Mining plan	Technical Committee observations	Within 15 days of receipt; observations raised only twice	MP Guidelines 2025
Mining plan	Proponent compliance	15 days, extendable to 30 days	MP Guidelines 2025
Mining plan	Disposal / in-principle / final	30 days / 5 working days / 3 days after payments confirmed	MP Guidelines 2025
Mining plan	Escrow for mine closure	Before mine opening; annual deposit by 30 September	MP Guidelines 2025
CBA Act	s.4(3) entry into dwellings	7 days' written notice to occupier	CBA s.4 proviso
CBA Act	s.7(1) notice to acquire	Within 2 years of s.4(1) (+1 year max); else s.4 lapses at 3 years	CBA s.7
CBA Act	s.8 objections	Within 30 days of s.7 notification	CBA s.8(1)
CBA Act	s.9(1) declaration	Within 3 years of s.7(1) notification	CBA s.9 proviso
CBA Act	Appeal against Tribunal award	Within 30 days to High Court	CBA s.20
EC	ToR	Within 60 days of Form-1, else applicant's ToR deemed approved	EIA Notif. para 7(i)
EC	Public hearing	Notice ≥30 days; hearing completed within 45 days of request	EIA Notif. App. IV
EC	Appraisal and decision	EAC within 60 days of final EIA; MoEFCC within 45 days (105 days)	EIA Notif. para 8
FC	State desks	DCF/Collector 20; inspections 20; Nodal/PCCF 15; PSC 30; State 15 working days	VSSR 2023 Schedule
FC	MoEFCC desks	Scrutiny 4; processing 5; IRO inspection 20; Advisory Committee 20; approval 10; letter 1 wd	VSSR 2023 Schedule
Rail (GCT)	Survey, IPA, ESP	Joint survey 15 days; IPA 20 days; ESP ≤4 weeks after IPA, Division 2 + HQ 2 weeks	GCT Circular 2026
Rail (GCT)	Construction	Start within 6 months; complete within 24 months of approval	GCT Circular 2026
Mine opening	Notice to DGMS, CCO, DM	To reach at least one month before commencement	Mines Act s.16(2)
Mine opening	Intimation to CCO after Board approval	Within 15 days of opening	CCR (Amendment) Rules 2026

Desk-wise limits are the PMC's best escalation tool: a file idle beyond its published limit can be raised with the next level by citing the rule.

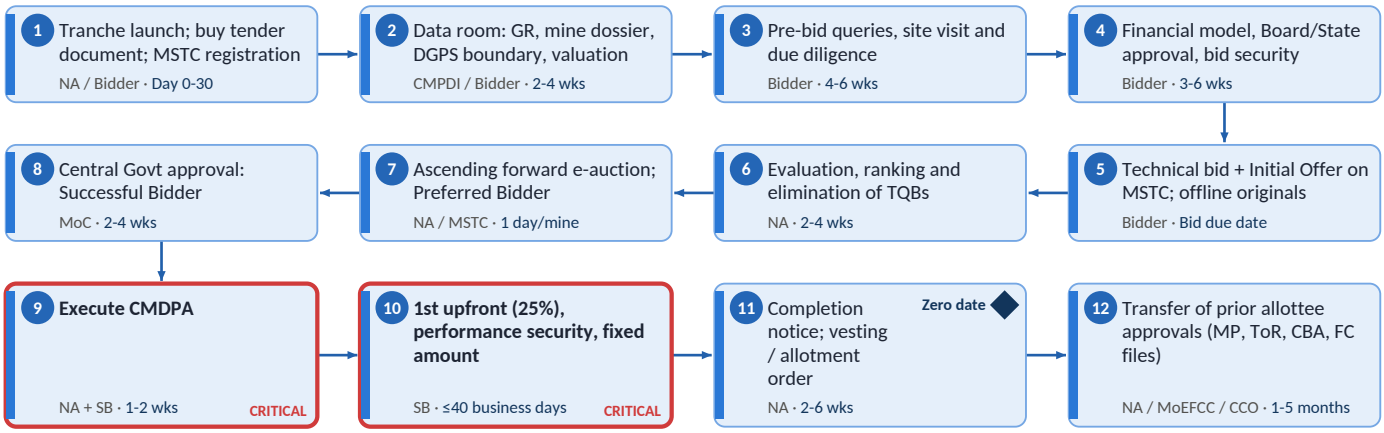
EVERY ACTIVITY, HOWEVER SMALL

5 Detailed process flows and step lists

Each sub-section gives the purpose and legal basis, a flowchart of the main steps (actor and typical time on each card; red outline = on the critical path of the generic network) and a granular checklist that can be lifted straight into the PMC tracker.

5.1 Pre-bid, e-auction, CMDPA and vesting

Commercial coal blocks are auctioned by the Nominated Authority (NA), Ministry of Coal, under the Coal Mines (Special Provisions) Act, 2015 and the MMDR Act, 1957, on the MSTC e-platform. The bid parameter is the **percentage share of revenue** payable to the State (floor 4%; increments of 0.5% up to 10% and 0.25% thereafter). There is no end-use restriction, no technical or financial eligibility threshold and 100% FDI is allowed under the automatic route. A two-stage process applies: technical bid plus initial offer, then an ascending forward e-auction among qualified bidders. Baitarani West was won by GMDC at a **Final Offer of 33.50%** (16th tranche under CM(SP) Act; CMDPA signed 29-Mar-2023; vesting order NA-104/4/2023-NA dated 08-Jun-2023).



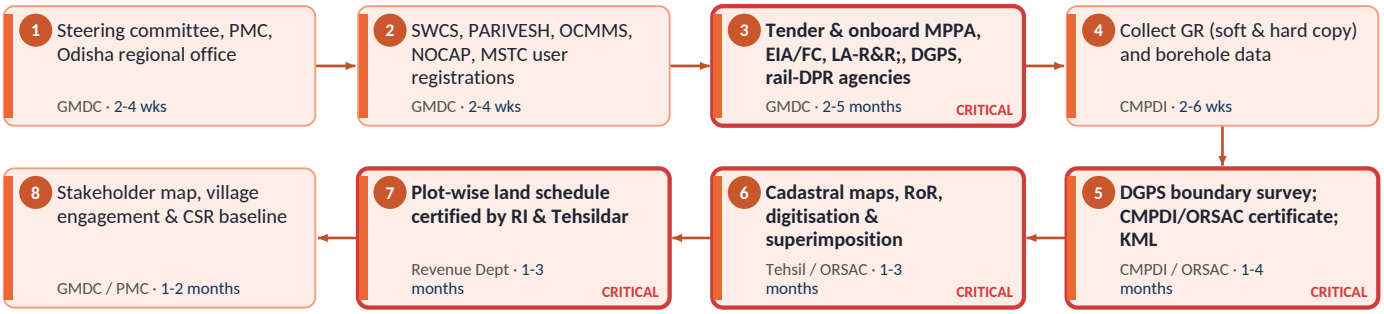
#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Track tranche notices; shortlist blocks on reserves, grade, strip ratio, land/forest share, logistics	Bidder (strategy)	MoC tranche notice	Continuous
2	Register on MSTC with DSC; purchase tender document	Bidder	Tender Document	Before cut-off
3	Data-room review: Geological Report, mine dossier, drone survey, valuation, prior-allottee data	Bidder / CMPDI	Tender Document	2-4 weeks
4	Submit written queries; attend pre-bid meeting; track corrigenda	Bidder / NA	Tender schedule	Per schedule
5	Request and carry out site visit (villages, forest patches, nalas, roads, rail heads, power, water)	Bidder	Tender schedule	1-2 weeks
6	Legal & social due diligence: land schedule split, Fifth Schedule/PESA status, ESZ & elephant corridors within 10 km, R&R; magnitude	Bidder / advisors	PESA 1996, WPA 1972	2-4 weeks
7	Market & logistics study: NCI-linked price, customers, evacuation route & cost	Bidder (marketing)	National Coal Index	2-4 weeks
8	Financial model: capex, MDO rates, revenue share, royalty 14% ad valorem, DMF 10% of royalty, NMET 2%	Bidder (finance)	MMDR s.9, 9B, 9C	2-3 weeks
9	Board approval (and State Govt approval for a State PSU) of bid ceiling and funding	Board / State Govt	Companies Act 2013	1-3 weeks
10	Bid security: 20% of upfront amount (fully explored) as BG/RTGS; 210-day validity + 60-day claim	Bidder	Tender Document	1 week
11	Prepare technical bid: eligibility affidavits, undertakings, POA, affiliate declarations	Bidder	Tender Document	2 weeks
12	Online submission on MSTC (technical + Initial Offer); offline originals to NA	Bidder	Annexure III cl. 3A	Bid due date
13	Technical evaluation; responsiveness; Technically Qualified Bidders declared	NA	Tender Document	2-3 weeks
14	Open Initial Offers; rank; eliminate (4-6 TQBs: lowest out; ≥7: lowest 1/3, max 3)	NA	Tender Document	1 week
15	Ascending forward e-auction; highest Final Offer = Preferred Bidder	MSTC / NA	Tender Document	1 day
16	NA recommends; Central Govt declares Successful Bidder; bid securities of others returned	MoC / NA	CM(SP) Rules	2-4 weeks
17	Execute CMDPA (Coal Mine Development and Production Agreement)	NA + SB	CM(SP) Act s.8	1-2 weeks
18	Pay 1st upfront instalment (25%) within 40 business days of CMDPA	SB	CMDPA cl. 5.2.1	≤40 business days
19	Furnish performance security: 65% of (1-yr royalty + 1-yr revenue share) at PRC	SB	CMDPA cl. 6.1.2	With completion notice
20	Pay Fixed Amount (mine dossier, prior-allottee land, GR and clearance costs)	SB	CMDPA cl. 3.1(d)	With completion notice
21	Submit Schedule B intimation and Schedule C completion notice with evidence	SB	CMDPA cl. 3.2.1	1 week

#	Activity / step	Responsible	Legal basis / reference	Typical time
22	Vesting / allotment order issued: efficiency-parameter clock (51 months) starts	NA	CM(SP) s.8	2-6 weeks
23	Apply for transfer of prior allottee's mining plan, ToR/EC, FC proposal, CBA notifications	SB	EIA Notif. para 11; VSSR 2023	1-5 months

- *Budget the full cash call before bidding: 1st upfront + PBG + fixed amount fall due within ~2 months of CMDPA.*
- *Price the R&R; burden and CA-land search into the bid: they, not geology, usually decide the 51-month timeline.*

5.2 Mobilisation, surveys and project set-up

The first 3-6 months after vesting decide whether every downstream clearance can start on time. The certified DGPS boundary and a plot-wise land schedule (khata, plot, kisam, area, owner) certified by the Revenue Inspector and Tehsildar are inputs to the mining plan, the FC proposal (KML on PARIVESH), the CBA notifications and the SES. Consultants are best tendered in parallel immediately after CMDPA signing.

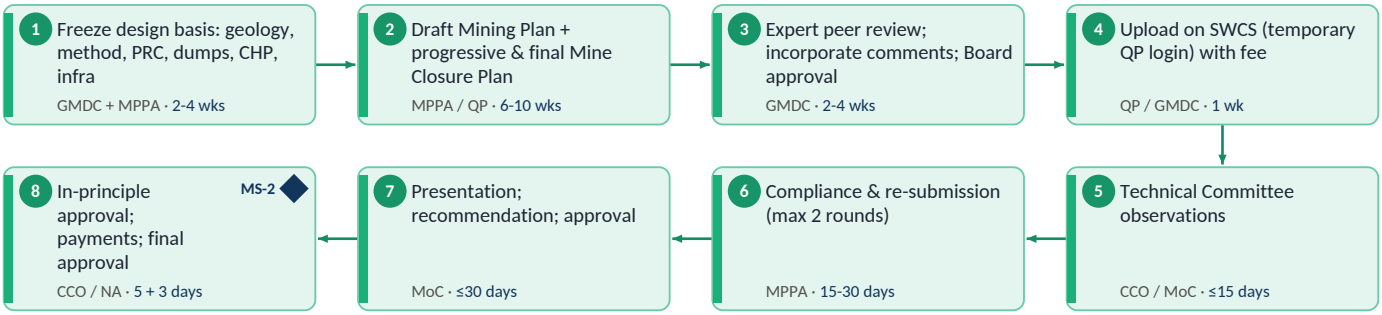


#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Constitute project steering committee; nominate project head and authorised signatories	GMDC	Board resolution	1-2 weeks
2	Appoint PMC; set up MIS, master tracker, risk register, monthly NA review pack	GMDC / PMC	-	2-4 weeks
3	Nominate authorised personnel; register on SWCS; obtain login credentials	GMDC / MoC	SWCS	2-4 weeks
4	Register as user agency on PARIVESH 2.0; OSPCB OCMMS; CGWA NOCAP; Odisha GO-SWIFT	GMDC	Portals	1-2 weeks
5	Approve organogram; hire GM (Project), DGM (Coal), land/R&R;, environment & forest officers	GMDC HR	-	3-6 months
6	Office space and guest house at Bhubaneswar and site; vehicles; IT connectivity	GMDC Admin	-	1-3 months
7	Tender: Mining Plan agency (accredited MPPA / Qualified Person)	GMDC	MCR 2016 r.22B	6-10 weeks
8	Tender: EC + FC consultant (NABET/QCI accredited for coal mining, Category A)	GMDC	EIA Notif. 2006	6-10 weeks
9	Tender: LA & R&R; agency (SES, genealogy, YADAST, case files, valuation)	GMDC	-	6-10 weeks
10	Tender: rail siding / road feasibility and DPR (e.g., IPRCL); transaction advisor for MDO	GMDC	GCT Policy	6-10 weeks
11	Obtain approved GR soft copy and hard copy, borehole logs, seam data from CMPDI	CMPDI	Tender Document	2-6 weeks
12	DGPS survey of block boundary; boundary pillars; certification by CMPDI; KML file	CMPDI / ORSAC	MP Guidelines 2025	1-4 months
13	Collect cadastral (village) maps, Record of Rights, settlement records	Tehsil office	OLR Act	1-2 months
14	Digitise maps; superimpose ML boundary on cadastral, forest (FMIS) and satellite layers	ORSAC / consultant	-	1-2 months
15	Prepare plot-wise land schedule: tenancy / Govt non-forest / forest; certify by RI & Tehsildar	Revenue Dept	-	1-3 months
16	Identify grazing (gochar), homestead, religious and community land needing replacement	Tehsildar / GMDC	OGLS Act	1 month
17	Stakeholder mapping; village meetings; CSR needs assessment and baseline	GMDC / PMC	Companies Act s.135	1-2 months
18	Logistics pre-feasibility (rail/road/port) and customer discovery	GMDC Marketing	-	2-3 months

- *Get the land schedule right first time - GMDC's Burapahar block had to revise its certified schedule after an ML-boundary change, holding up s.9(1).*
- *Insist that every consultant contract has milestone-linked payments and liquidated damages tied to CMDPA dates.*

5.3 Mining Plan and Mine Closure Plan (MS-2)

Under the Mining Plan Guidelines for Coal and Lignite Blocks 2025 (MoC, Jan-2025) the plan is prepared by a Qualified Person / accredited MPPA, approved by the company Board, peer reviewed and filed on SWCS with the fee. The Technical Committee must raise observations within 15 days (only twice); the proponent replies within 15 days (max 30); the approving authority disposes within 30 days of (re)submission; CCO conveys in-principle approval within 5 working days and final approval within 3 days of payments being confirmed by the NA. MS-2 is due 6 months from vesting. For Baitarani West the OMC plan approved on 10-Nov-2021 was deemed transferred with the vesting order.

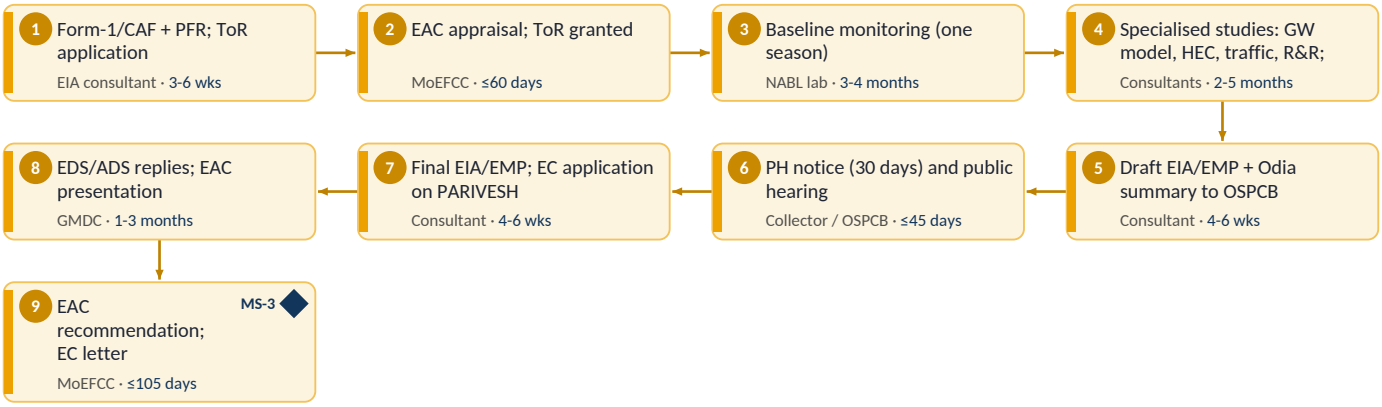


#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Finalise project assumptions (geology, mine design, method, PRC, mine life) - management sign-off	GMDC + MPPA	-	2-4 weeks
2	Mine design: pit limits, stripping ratio, calendar plan, dump design, CHP, drainage, infrastructure	MPPA	MP Guidelines 2025	4-8 weeks
3	Mine Closure Plan (progressive + final) with cost, ecological restoration and post-mining land use	MPPA	MCP Guidelines	2-4 weeks
4	Attach CMPDI DGPS certificate, KML and approved GR; QP certifies plan within block boundary	QP	MP Guidelines 2025	-
5	Submit draft to GMDC; expert reviewer comments; pre-check; re-share; go-ahead	MPPA / Expert	Tracker MP4-MP9	3-6 weeks
6	Company Board approval of the Mining Plan	GMDC Board	MP Guidelines 2025	1-2 weeks
7	Upload through temporary login on SWCS; pay non-refundable fee; submit	QP / GMDC	SWCS	1 week
8	Technical Committee scrutiny; observations uploaded (twice only)	CCO / Tech. Committee	MP Guidelines 2025	≤15 days
9	Address observations; re-submit modified plan	MPPA / GMDC	MP Guidelines 2025	15 days (max 30)
10	Presentation before committee; recommendation for approval	GMDC / MoC	-	1-2 weeks
11	Approving authority disposes the application	MoC	MP Guidelines 2025	≤30 days
12	CCO conveys in-principle approval to proponent and NA	CCO	MP Guidelines 2025	≤5 working days
13	Revise performance security if PRC changed; pay dues; NA confirmation	GMDC / NA	CMDPA cl. 6.1.4(c)	≤30 days
14	CCO conveys final approval (MS-2 achieved)	CCO	CMDPA Sch. D	≤3 days
15	Later revisions: minor changes (e.g., ≤50% capacity increase, HEMM change) by Board; others to Coal Controller	GMDC	MP Guidelines 2025 para 2.9	As needed

- *A plan that front-loads production pays: coal produced before the Scheduled Date of Production gets a 50% rebate on the revenue share.*
- *Keep the box-cut in a no-habitation patch (Govt/acquired land) so mine opening is not hostage to R&R; shifting.*

5.4 Environment Clearance (MS-3)

A coal mine is Category A (item 1(a)) under the EIA Notification 2006 and is appraised by the Expert Appraisal Committee (Coal Mining) of MoEFCC on PARIVESH. ToR is due within 60 days of Form-1/CAF (standard ToR may issue faster); baseline covers one season; the public hearing must be completed within 45 days of the request; EC is to be decided within 105 days of the final EIA. MS-3 (EC) is due 18 months after MS-2. Baitarani West: ToR transferred 06-Nov-23, PH 24-Sep-24, EAC returned the proposal on 03-Apr-25, elephant-corridor NOC 25-Aug-25, fresh EC application 30-Sep-25, EC granted 03-Dec-25.

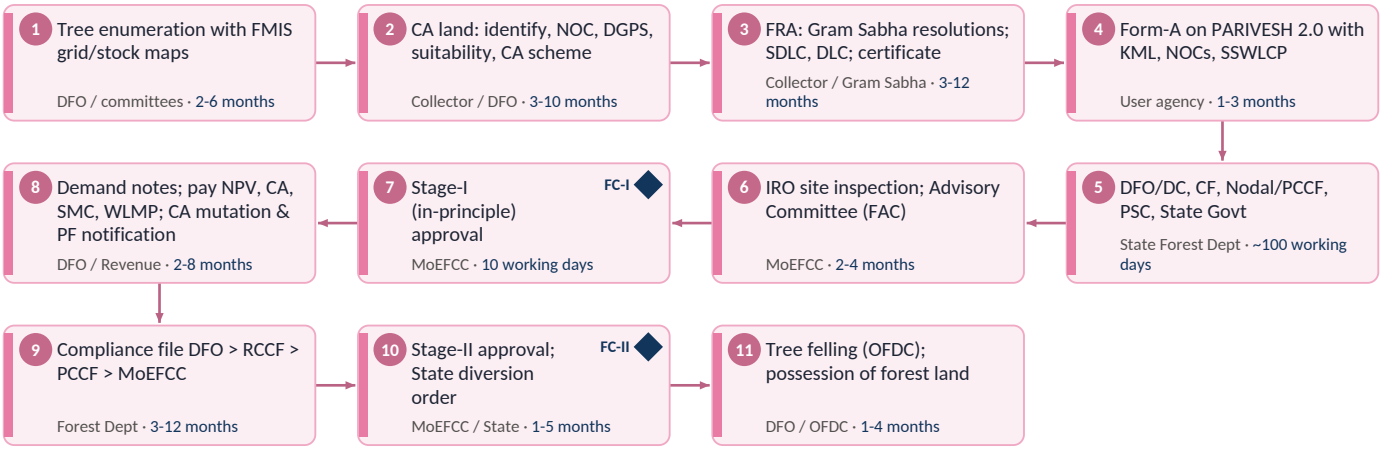


#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Prepare Pre-Feasibility Report and Form-1 / Common Application Form; KML; approved mine plan	EIA consultant	EIA Notif. App. I	3-6 weeks
2	Online ToR application on PARIVESH (or ToR transfer in Form-8 for a prior allottee's ToR)	GMDC	EIA Notif. para 11	1 week
3	ToR presentation before EAC (Coal Mining); standard/specific ToR granted	MoEFCC / EAC	EIA Notif. para 7(ii)	≤60 days
4	Accept ToR or seek amendment (e.g., after mining plan revision)	GMDC	-	2-4 weeks
5	Baseline data: AAQ, water, noise, soil, ecology & biodiversity, traffic, socio-economics (one season)	NABL lab / consultant	Std. ToR	3-4 months
6	Hydrogeological investigation and groundwater modelling; mine dewatering estimate	Consultant	ToR / CGWA	2-4 months
7	Human-elephant conflict & wildlife study; site-specific wildlife conservation plan input	Consultant / WII-type agency	ToR	2-4 months
8	Traffic study, subsidence/slope stability, cumulative impact (Talcher coalfield), R&R; plan	Consultants	ToR	2-3 months
9	Prepare draft EIA/EMP and executive summary in English and Odia	Consultant	EIA Notif. App. III	4-6 weeks
10	Submit to Member Secretary, OSPCB requesting public hearing; fee	GMDC	EIA App. IV	1 week
11	Publish PH notice in two newspapers (one Odia) - at least 30 days before hearing	OSPCB	EIA App. IV para 3	30 days
12	Conduct public hearing (DM/ADM presides; video-recorded); proceedings in 8 days	Collector / OSPCB	EIA App. IV	≤45 days
13	OSPCB forwards PH proceedings to MoEFCC and Regional Office	OSPCB	EIA App. IV	1-2 weeks
14	Incorporate PH issues & action plan with budget into final EIA/EMP	Consultant	-	3-6 weeks
15	Submit final EIA/EMP and EC application; respond to EDS/ADS	GMDC	EIA Notif. para 7(iv)	2-8 weeks
16	Final technical presentation before EAC; EAC recommendation (or return / deferral)	EAC	EIA Notif. para 7(iv)	≤60 days
17	MoEFCC decision and EC letter with specific conditions	MoEFCC	EIA Notif. para 8	≤45 days after EAC
18	Upload six-monthly compliance reports; display EC; publish in newspapers	GMDC	EC conditions	Ongoing

- *BW lesson: EAC returned the first proposal (Apr-25); build the elephant-corridor study, R&R; plan and FC status into the EIA from day one.*
- *Run baseline in the first non-monsoon season after vesting; a missed season costs ~6 months.*

5.5 Forest clearance, FRA, CA land, tree enumeration and wildlife plan

Diversion of forest land (BW: 64.37 ha = 4.47 ha reserved + 59.90 ha revenue forest) needs prior approval under the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 and Rules 2023 (VSSR), processed on PARIVESH 2.0. VSSR prescribes working-day limits at every desk (DCF/Collector 20, site inspections 20, Nodal/PCCF 15, PSC 30, State 15, MoEFCC scrutiny/IRO/Advisory Committee 4+5+20+20, approval 10). Stage-I (in-principle) is followed by payment of compensatory levies (NPV, CA, SMC, wildlife plan), compliance of conditions and settlement of rights under FRA 2006 before Stage-II and the State's diversion order. Four sub-processes run in parallel: FRA, CA land, tree enumeration and the site-specific wildlife conservation plan.

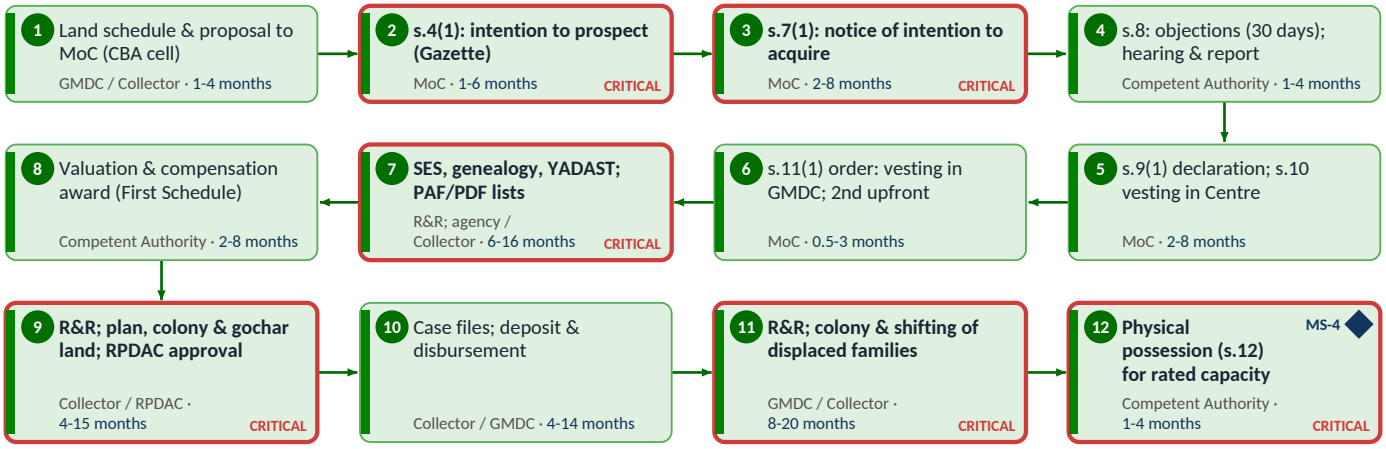


#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Letter from DFO to FMIS for grid and stock maps; demand note; pay fees; maps issued	DFO / FMIS	Tracker TE1-TE4	3-6 weeks
2	Form tree-enumeration committees (RO); enumerate; report to DFO; DFO approval	Range Officer / DFO	Tracker TE5-TE10	1-4 months
3	Identify CA land: equivalent non-forest Govt land (or 2x degraded forest where permitted)	Collector / GMDC	VSSR 2023 r.11	1-4 months
4	Collector NOC for CA land; Gram Sabha in CA villages; DGPS of CA land; DSS check	Collector / DFO	Tracker CA1-CA6	1-3 months
5	Suitability certificate from DFO/CCF; prepare and get CA scheme technically vetted	DFO / CF	Tracker CA7-CA9	1-2 months
6	Transfer & mutation of CA land to Forest Dept; notify as PF/RF	Tehsildar / Collector	Tracker CA10-CA13	3-6 months
7	FRA: application to Collector; SDM starts process; Gram Sabha resolutions (quorum, minutes, video)	Collector / Gram Sabha	FRA 2006 s.6; Rules r.4	1-3 months
8	FRC claims verification; SDLC and DLC recommendations; FRA certificate by Collector	SDLC / DLC	FRA Rules r.12	2-6 months
9	Prepare site-specific wildlife conservation plan (elephant corridor measures); CWLW approval	Consultant / CWLW	WPA 1972	3-6 months
10	Obtain NOCs: Gram Panchayat, Archaeology, Collector; details of land and R&R;	GMDC	VSSR checklist	1-3 months
11	File Form-A (Part I) on PARIVESH 2.0 with KML, maps, mine plan, CA details	User agency	VSSR 2023 r.9	2-4 weeks
12	Nodal Officer completeness check; EDS loop	State Nodal Officer	VSSR timelines	≤10 working days
13	DFO/DC processing; site inspection by DCF/CF; CF recommendation	DFO / CF	VSSR Schedule	40 working days
14	Nodal Officer/PCCF processing; Project Screening Committee; State Govt recommendation	PCCF / PSC / State	VSSR Schedule	60 working days
15	MoEFCC scrutiny; IRO site inspection & report; Advisory Committee (FAC)	MoEFCC / IRO / FAC	VSSR Schedule	~50 working days
16	Stage-I (in-principle) approval with conditions	MoEFCC	Adhiniyam s.2	10 working days
17	DFO raises demand: NPV, CA cost, SMC plan, wildlife plan, additional CA norms; pay on e-portal	DFO / GMDC	VSSR 2023	1-3 months
18	Prepare & get approved SMC plan, avifauna & other schemes (BW: 11 of 12 approved 10-Apr-26)	DFO / RCCF	FC-I conditions	2-6 months
19	Compliance report: DFO > RCCF > PCCF (Nodal) > Secretary (Forest) > MoEFCC	Forest Dept	VSSR 2023	2-6 months
20	Stage-II (final) approval; State Govt order for diversion / lease	MoEFCC / State	VSSR 2023 r.9-10	1-3 months
21	Tree-felling application; marking; APCCF approval; demand for felling cost; payment	DFO / APCCF	Tracker TFPP1-TFPP12	1-3 months
22	Felling, debarking, logging, transport by OFDC; possession of felled area with coordinates	OFDC / DFO	Tracker TFPP13-TFPP19	1-4 months

- *CA land is the classic long pole: BW's CA land is in Koraput district (Baipariguda), 300+ km from the mine; start the search at the bid stage.*
- *FRA certificate is a hard prerequisite for the State diversion order - BW's is still pending (Oct-26).*

5.6 Land acquisition under the CBA Act, 1957 and R&R; (MS-4)

For a Government company, land is acquired by the Central Government under the Coal Bearing Areas (Acquisition & Development) Act, 1957 and vested in the company by a s.11(1) order; that order substitutes for a mining lease for CMDPA purposes (2nd upfront instalment is due within 15 business days). Under the RFCTLARR (Removal of Difficulties) Order 2015 (S.O. 2368(E), 28-Aug-2015) the **First, Second and Third Schedules of RFCTLARR 2013** (compensation, R&R;, infrastructure amenities) apply to CBA acquisitions; Odisha's R&R; Policy 2006 (RPDAC) and ORFCTLARR Rules 2016 govern implementation. Statutory clocks: s.7(1) within 2 years of s.4(1) (+1 year); objections within 30 days of s.7; s.9(1) declaration within 3 years of s.7(1). MS-4 (possession and R&R; for rated capacity) is due 21 months after MS-3.



#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Prepare land documents for CBA: schedule, maps, khatian, RoR, area statement	GMDC / LA agency	CBA s.4(2)	1-2 months
2	Proposal for s.4(1) to MoC through Coal Controller / CBA cell; vetting; Gazette notification	MoC	CBA s.4(1)	1-6 months
3	Survey, bore, mark boundaries under s.4(3); pay/tender damages under s.6	Competent Authority	CBA s.4(3), s.6	During validity
4	Notice of intention to acquire land/rights - within 2 years of s.4(1) (+1 year max)	MoC	CBA s.7(1)-(2)	2-8 months
5	Receive objections within 30 days; hearing in person/by counsel; report to Central Govt	Competent Authority (Collector/ADM)	CBA s.8	1-4 months
6	Declaration of acquisition (parcel-wise allowed) - within 3 years of s.7(1); copy to State	MoC	CBA s.9(1)-(2)	2-8 months
7	Urgency route if justified: s.8 dispensed, declaration any time after s.7	MoC	CBA s.9A	Optional
8	Vesting in Central Govt free of encumbrances on publication	MoC	CBA s.10	Automatic
9	Order directing vesting in GMDC (Govt company) on terms; pay 2nd upfront within 15 business days	MoC / GMDC	CBA s.11(1); CMDPA 5.2.1	0.5-3 months
10	Socio-economic survey (SES) of all affected families; cut-off date; PAF/PDF lists	R&R; agency / Collector	RFCTLARR Sch. II; ORRP 2006	4-10 months
11	Genealogy (family tree) and YADAST (enumeration of houses, structures, trees, wells)	R&R; agency / RI	ORFCTLARR Rules 2016	4-10 months
12	Valuation: market value x multiplier (rural factor), 100% solatium, assets; 12% additional amount	Competent Authority	RFCTLARR Sch. I	2-6 months
13	Compensation determined; disputes to Tribunal; appeal to High Court in 30 days	CA / Tribunal	CBA s.13, 14, 20	2-8 months
14	Draft R&R; plan: homestead, house, subsistence, employment/annuity, shifting, cattle shed, artisans	GMDC / Collector	RFCTLARR Sch. II; ORRP 2006	2-4 months
15	Identify R&R; colony land and replacement gochar land; Third-Schedule amenities	Collector / GMDC	RFCTLARR Sch. III	2-6 months
16	RPDAC meeting (Collector chairs; MLA, MP, PRI, PAF reps); R&R; plan approval; Govt notification	Collector / RDC	ORRP 2006 para 6	2-8 months
17	Prepare & validate case files (title, mutation, bank details); deposit with Competent Authority	LA agency / Collector	CBA s.17	2-6 months
18	Disbursement camps; tender/deposit compensation; deposit with Tribunal if disputed	Collector	CBA s.17	3-9 months
19	Government land: alienation/premium or transfer for land in ML area; possession letter	Revenue & DM Dept	OGLS Act / CBA s.9 proviso	3-12 months
20	R&R; colony development: plots, roads, water, power, school, health post, community hall	GMDC / contractor	RFCTLARR Sch. III	8-18 months
21	Shifting of displaced families (phase-wise, initial mining area first); shifting allowance	GMDC / Collector	ORRP 2006	2-6 months

#	Activity / step	Responsible	Legal basis / reference	Typical time
22	Notice to surrender possession; take possession; demarcate and fence	Competent Authority	CBA s.12	1-3 months
23	Constitute project-level R&R; monitoring; grievance redressal; periphery development	Collector / GMDC	ORRP 2006	Ongoing

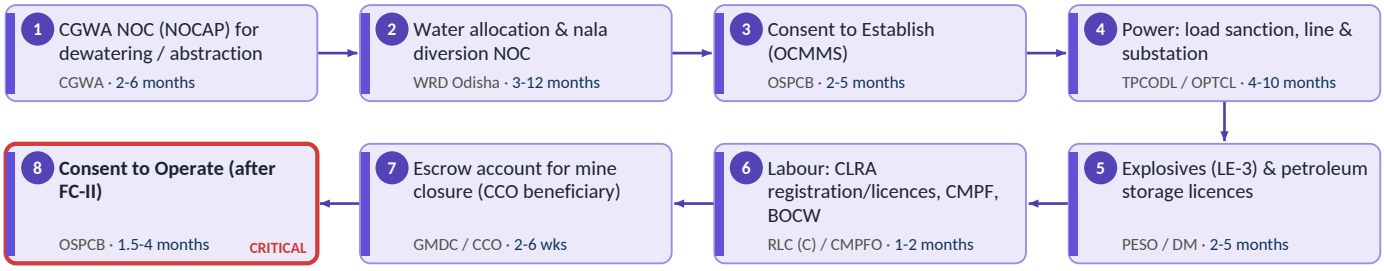
Entitlements that flow to affected families (RFCTLARR 2013 Schedules applied to CBA acquisitions by S.O. 2368(E), 28-Aug-2015):

Head	Entitlement
Compensation (First Schedule)	Market value x multiplication factor notified by the State (1-2 in rural areas) + value of structures, trees and assets + 100% solatium; 12% p.a. additional amount from notification to award/possession
Housing (Sch. II)	Displaced family: house as per Indira Awas Yojana specification in rural areas (or its cost if declined)
Employment / annuity	Employment to one member per affected family, or one-time Rs 5 lakh, or annuity of not less than Rs 2,000/month for 20 years (indexed)
Subsistence grant	Rs 3,000 per month for one year to each displaced family; additional Rs 50,000 for SC/ST families displaced from Scheduled Areas
Transport & resettlement	One-time Rs 50,000 transportation and Rs 50,000 resettlement allowance
Livelihood support	Cattle shed / petty shop and artisan / small trader: one-time assistance of at least Rs 25,000 each
Charges	Stamp duty and registration fee for land and houses allotted are borne by the requiring body
Third Schedule amenities	Roads, drainage, drinking water, school, anganwadi, health centre, panchayat ghar, PDS outlet, places of worship, cremation ground, electricity, etc. in the R&R; colony
Odisha overlay	ORRP 2006 / RPDAC-approved package; the higher of the State package and the Second Schedule should be applied - confirm with the Collector

- *SES must capture actual families, not only recorded tenants - BW's SES (3,402 original families, Jan-25) created an issue at RPDAC.*
- *Monthly rate revisions force re-work of case files; batch disbursement by village and lock rates via the award date.*

5.7 Other statutory consents, permissions and registrations

CMDPA Schedule D lists many 'non-monitored' activities that are nevertheless pre-conditions for opening: nala/river diversion, diversion of power lines/roads, permission to draw water and power, CTE/CTO, DGMS intimation, explosives and petroleum licences, Factories Act permission, radio-frequency licence, labour permissions and the escrow account. Each has its own portal and clock; most can and should run in parallel with EC/FC so that only FC Stage-II remains when the CTO is due.



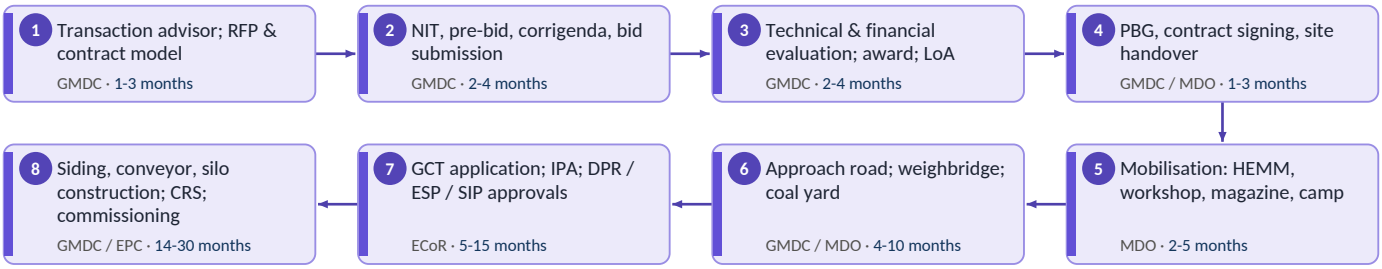
#	Activity / step	Responsible	Legal basis / reference	Typical time
1	CGWA NOC for mine dewatering & GW abstraction on NOCAP (impact assessment, piezometers, recharge plan)	CGWA	CGWA Guidelines 2020	2-6 months
2	Surface-water allocation for dust suppression/CHP/colony; agreement with WRD	WRD Odisha	Odisha Irrigation Act	3-9 months
3	Nala/stream diversion: DPR, joint field survey (WRD, Minor Irrigation), NOC (BW: due 30-Oct-26)	WRD / Chief Engineer	CMDPA Sch. D	4-12 months
4	Diversion of power line / road / canal within lease, if any	OPTCL / R&B; / WRD	CMDPA Sch. D	3-9 months
5	CTE: online application; hard copy to Member Secretary; RO inspection; technical committee; grant	OSPCB	Water Act s.25; Air Act s.21	2-5 months
6	Hazardous waste authorisation (used oil, batteries); e-waste registration	OSPCB	HOWM Rules 2016	1-2 months
7	Power: application, feasibility, estimate, deposit; 33 kV line / substation construction	TPCODL / OPTCL	Electricity Act 2003	4-10 months
8	Explosives: magazine site approval, DM NOC, licence in Form LE-3; shot-firer/blaster competency	PESO / DM	Explosives Rules 2008	2-5 months
9	Petroleum (HSD) storage licence for mine fuel station	PESO	Petroleum Rules 2002	2-4 months
10	Factories Act licence for CHP / workshop where applicable	Director of Factories	Factories Act 1948	1-2 months
11	WPC licence for radio-frequency communication (VHF/UHF sets)	DoT / WPC	Indian Telegraph Act	1-3 months
12	Principal employer registration (CLRA) and contractor licences - Central sphere	RLC (Central)	CLRA 1970	1 month
13	CMPF registration of mine & employees; BOCW registration for construction	CMPFO / Labour Dept	CMPF Act 1948	1 month
14	Fire NOC, building plan approvals (site office, colony), legal-metrology stamping of weighbridges	Local bodies / LM Dept	Various	1-3 months
15	Escrow account (FD with CCO as exclusive beneficiary) at Rs 14 lakh/ha OC base rate (May-24, WPI-escalated)	GMDC / CCO	MP Guidelines 2025	2-6 weeks

#	Activity / step	Responsible	Legal basis / reference	Typical time
16	CTO: application with EC, FC-II, CTE compliance; queries; inspection; grant (BW applied 08-Jul-26)	OSPCB	Water & Air Acts	1.5-4 months

- *Apply for CTO as soon as CTE conditions are met; keep only FC-II outstanding (BW followed this approach).*
- *Nala diversion often needs Gram Sabha and WRD field verification - BW's DPR (Mar-26) is still awaiting NOC.*

5.8 Mine developer (MDO), infrastructure and evacuation

GMDC operates through a turnkey contractor/MDO for coal and OB. Rail is the binding long-term constraint: the EC requires 12 of 15 MTPA to move by in-pit conveyor, silo and a private siding connected through the common rail corridor to ECoR at Jarpara, and the siding must be ready before output reaches 6 MTPA. Under the Gati Shakti Cargo Terminal (GCT) Master Circular 2026: joint survey within 15 days, in-principle approval within 20 days, ESP approval in 4+2+2 weeks, signalling plan in 4 weeks, construction to start within 6 months and finish within 24 months. Early coal moves by covered trucks on the strengthened Chhendipada-Jharpada road (~25 km).

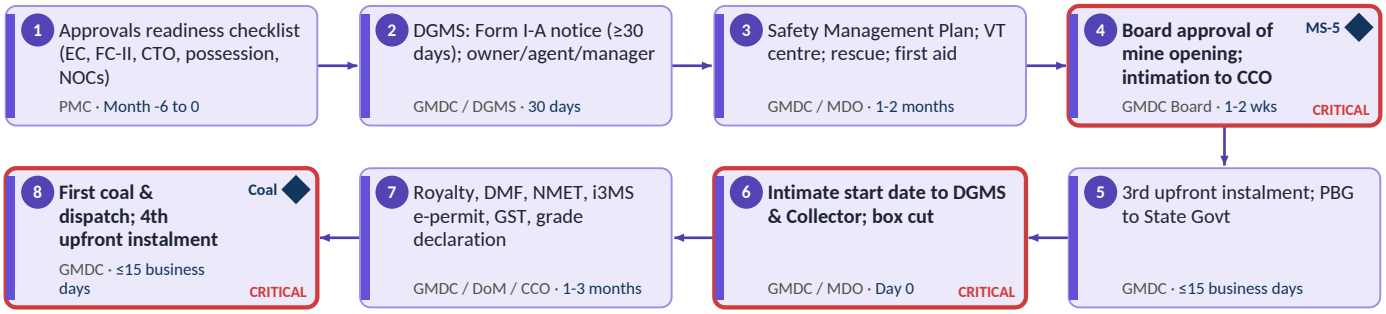


#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Appoint transaction advisor; choose model (MDO vs turnkey coal + OB contract); approvals	GMDC	Tracker 'Tender-MDO'	1-3 months
2	Prepare RFP: scope, HEMM, rates, minimum guarantee, LDs, safety, PBG	GMDC / TA	GFR / GMDC manual	1-2 months
3	Upload NIT; pre-bid meeting (BW: 17-Feb-25, 14 bidders); corrigenda	GMDC	-	1-2 months
4	Bid submission; technical evaluation; financial opening / reverse auction	GMDC	-	1-3 months
5	Competent-authority approval; LoA; PBG (BW: LoI 13-Nov-25, PBG 30-Nov-25)	GMDC / MDO	-	1-2 months
6	Sign contract; hand over site (BW: handover 10-Aug-26; contract 17-Sep-26)	GMDC / MDO	-	1-3 months
7	Mobilise HEMM (surface miners, shovels, dumpers), workshop, fuel station, camp, magazine	MDO	CMR 2017	2-5 months
8	Strengthen Chhendipada-Jharpada road (24.985 km incl. 3 bypasses); approach to ML boundary	GMDC / R&B;	EC condition	6-15 months
9	Interim dispatch: weighbridges, CCTV, RFID, coal yard, covered trucks	GMDC / MDO	EC; Legal Metrology	3-6 months
10	GCT application with fee (Rs 20,000 + GST); joint location survey; feasibility; IPA (BW IPA: 18-Jan-25)	Division (DRM)	GCT Circular 2026	~35 days
11	DPR/ESP submission within 4 weeks of IPA; Division 2 weeks; HQ (CE/P&D;) 2 weeks (BW: DPR 17-Jun-25)	ECoR	GCT Circular 2026	2-12 months
12	Signalling interlocking plan; land for siding & coal stock at Jharpada	ECoR / GMDC	GCT Circular 2026	1-6 months
13	Construct siding, in-pit conveyor, silo; CRS inspection; agreement (35 yrs); commercial notification	GMDC / EPC / ECoR	GCT Circular 2026	14-30 months

- *Rail is not on the path to first coal but becomes critical for ramp-up: plan its latest start from the 6 MTPA year.*
- *Lock the MDO's mobilisation to the forecast FC-II date to avoid idle-HEMM claims.*

5.9 Mine opening, first coal and production compliance (MS-5)

The Colliery Control (Amendment) Rules, 2025 (notified 23-Dec-2025) removed the Coal Controller's prior mine-opening permission: the Board of a company now approves opening of a mine/seam once all Central, State and statutory approvals are in hand, and intimates CCO (a further 2026 amendment specifies intimation within 15 days of opening). The 3rd upfront instalment is due within 15 business days of the opening permission and the performance security moves to the State Government. The notice of opening (Form I-A) must reach DGMS, the Coal Controller and the District Magistrate at least one month before operations begin (Mines Act s.16; CMR 2017 reg. 3).



#	Activity / step	Responsible	Legal basis / reference	Typical time
1	Approvals readiness review: EC, FC-II & State order, CTO, possession, CGWA, water, nala, power, licences	PMC / GMDC	CCR 2004 r.9 (amended)	Monthly from M-6
2	Appoint owner/agent/manager (First-Class Manager's certificate) and statutory officials; notices to DGMS	GMDC / MDO	Mines Act s.17; CMR 2017	1 month
3	Notice of opening in Form I-A to DGMS (Chief Inspector), Coal Controller & DM - ≥ 30 days ahead	GMDC	Mines Act s.16; CMR 2017 reg. 3	≥30 days
4	Safety Management Plan, risk assessment, traffic rules for HEMM, blasting SOP, dump-slope study	GMDC / MDO	CMR 2017	1-2 months
5	Vocational training centre, first-aid, rescue-station tie-up, crèche, canteen, rest shelters	GMDC / MDO	MVT Rules 1966; Mines Rescue Rules 1985	1-3 months
6	Board resolution approving opening of mine/seam (Companies Act company)	GMDC Board	CCR (Amendment) Rules 2025/2026	1-2 weeks
7	Intimation of opening to Coal Controller's Organisation (within 15 days)	GMDC	CCR (Amendment) Rules 2026	≤15 days
8	Pay 3rd upfront instalment (25%) to State Govt; submit performance security to State; NA returns its PBG	GMDC	CMDPA 5.2.1, 6.2	≤15 business days
9	Revise PBG (first revision on NCI at opening permission)	GMDC	CMDPA 6.1.4(a)	≤30 days
10	Intimate actual date of start to DGMS and Collector; box cut; OB removal	GMDC / MDO	Tracker MO7-MO9	Day 0
11	Register for royalty/DMF/NMET with Deputy Director of Mines (Talcher); i3MS e-transit permits	GMDC / DoM Odisha	MMDR s.9, 9B, 9C	1-2 months
12	Coal grade declaration (third-party sampling) and quality systems	CCO / GMDC	Colliery Control Rules	1-3 months
13	Sales: customer contracts / e-auction, pricing linked to NCI, invoicing, GST	GMDC Marketing	-	1-3 months
14	First coal dispatch; pay 4th upfront instalment within 15 business days	GMDC	CMDPA 5.2.1	≤15 business days
15	Monthly revenue-share payments (Final Offer x quantity x max(notional, actual price)); upfront adjusted up to 50%	GMDC	CMDPA cl. 9	Monthly
16	Monthly efficiency-parameter report to NA; annual production ≥ 65% (3-yr block ≥ 75%) of plan	GMDC	CMDPA cl. 10, 16	Monthly / annual
17	Annual escrow deposit by 30 Sept; six-monthly EC compliance; Form-wise DGMS returns	GMDC	MP Guidelines 2025; EC	Recurring

- *Mine-opening date drives three cash events (3rd upfront, PBG transfer, PBG revision) - provision them in the treasury plan.*
- *Every tonne before the Scheduled Date of Production (BW: 08-Sep-2029) attracts only half the 33.5% revenue share.*

HOW THE SCHEDULE WAS COMPUTED

6 PERT and CPM - method

The ~200 steps were rolled up into **62 activities** that each have one accountable owner and clear predecessors (activity-on-node, finish-to-start). Time zero (T0) is the launch of the auction tranche. For every activity three estimates were set from statutory time limits, the durations actually observed on Baitarani West and the MoC's July-2025 review of Odisha blocks:

Quantity	Formula	Meaning
Expected time t_e	$(a + 4m + b) / 6$	Beta-PERT mean of optimistic a, most likely m, pessimistic b
Variance σ^2	$((b - a) / 6)^2$	Uncertainty of the activity; summed along a path
Forward pass	ES = max(EF of predecessors); EF = ES + t_e	Earliest the activity can start / finish
Backward pass	LF = min(LS of successors); LS = LF − t_e	Latest without delaying the finish
Total float	TF = LS − ES	Slack; TF = 0 defines the critical path
Free float	FF = min(ES of successors) − EF	Slack that does not disturb any successor
Probability	$Z = (T_{target} - T_E) / \sqrt{\Sigma\sigma^2_{CP}}$	Classical PERT (critical path only)
Monte Carlo	20,000 beta-PERT samples per activity	Captures merge bias of near-critical paths; criticality index = share of runs in which an activity is critical

Worked example - L12 R&R colony and shifting: a = 8, m = 12, b = 20 months → t_e = (8 + 4×12 + 20) / 6 = **12.67** months; σ = (20 − 8) / 6 = 2.00. ES = EF of L9 = 42.9; EF = 55.6; LS = 42.9; TF = 0.0 → critical.

Key modelling assumptions

- Fully explored block (no GR to prepare), opencast, Government company buyer, so land vests through CBA s.11(1); a mining-lease route (private bidder) would replace L2-L6 with the State ML process.
- Mine opening (MS-5) follows full MS-4 (possession and R&R for the rated-capacity area) in the baseline, mirroring the CMDPA sequence; scenario S3 relaxes this to phased possession.
- EC appraisal waits for the FC proposal to be on PARIVESH (EAC routinely asks for FC status); CTO waits for EC-linked CTE, CGWA NOC and FC Stage-II.
- Rail siding is excluded from the first-coal network (initial road dispatch) and tested separately against the 6 MTPA year (Section 10).
- No resource constraints, no calendar effects (monsoon, elections) beyond what is embedded in the pessimistic values.

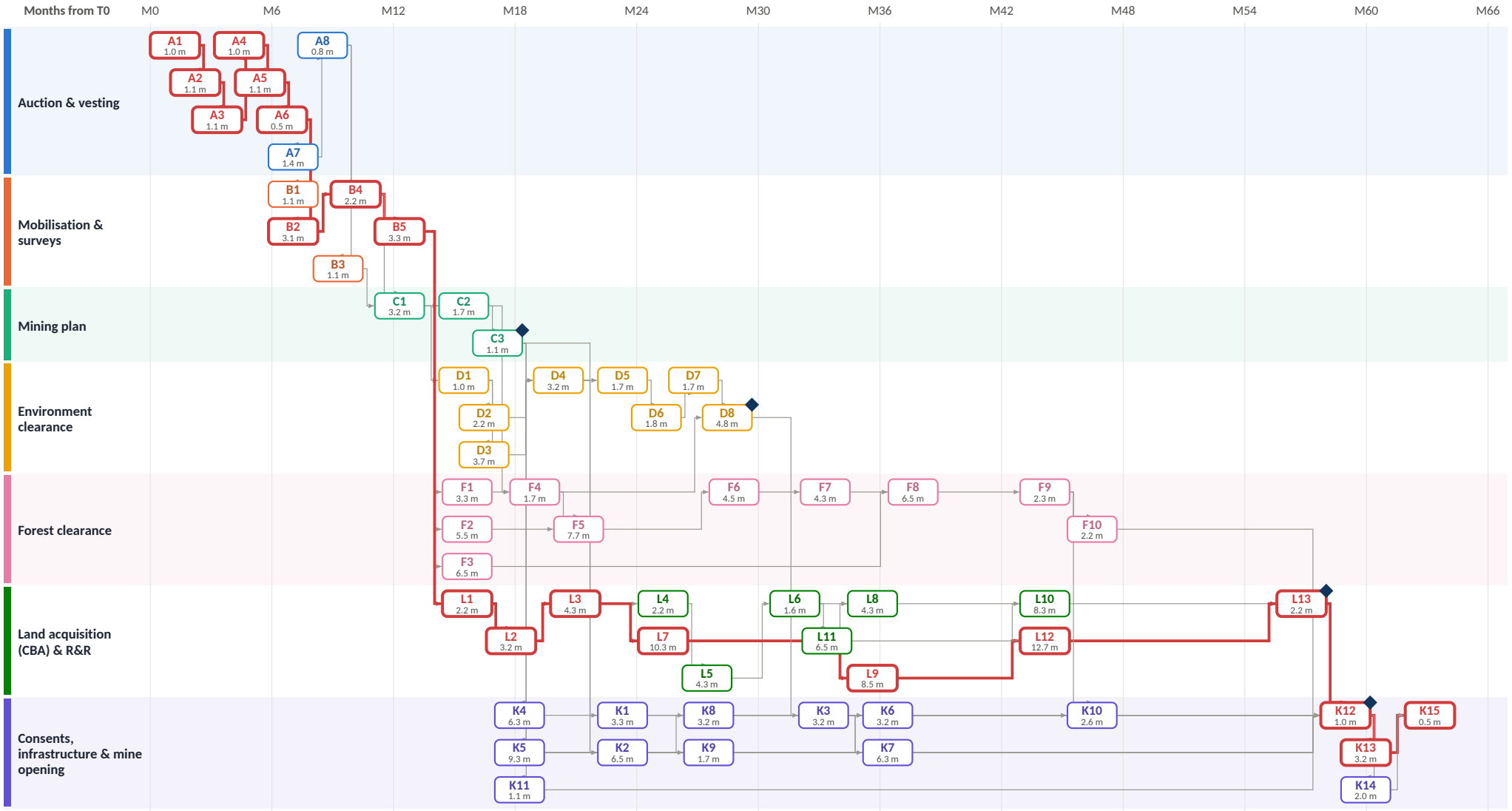
CPM RESULTS

7 Network, critical path and schedule

Expected project duration T_E = **62.5 months** from T0 to first coal. Critical path: **A1 → A2 → A3 → A4 → A5 → A6 → B2 → B4 → B5 → L1 → L2 → L3 → L7 → L9 → L12 → L13 → K12 → K13 → K15**. Variance along the path to MS-5 = 13.6 (σ = 3.7 months). Columns: a/m/b three-point estimates, t_e, σ, earliest/latest start and finish, total float (TF) and Monte Carlo criticality index (CI). Red rows have zero total float; CI ≥ 50% means the activity drives the finish in at least half the simulated futures.

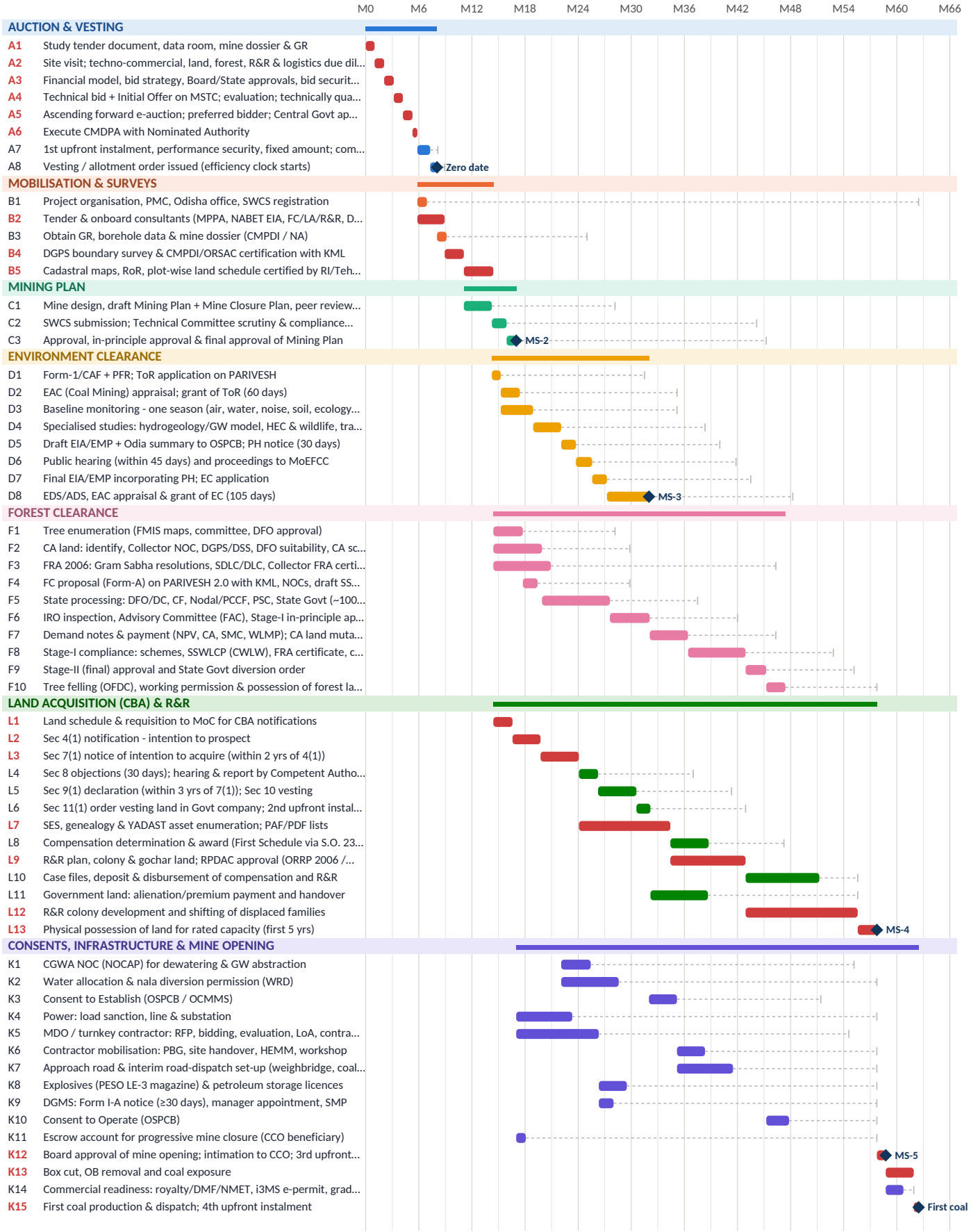
ID	Activity	a	m	b	te	σ	ES	EF	LS	LF	TF	CI
A1	Study tender document, data room, mine dossier & GR	0.5	1	1.5	1.0	0.2	0.0	1.0	-0.0	1.0	0.0	100%
A2	Site visit; techno-commercial, land, forest, R&R; & logistics...	0.5	1	2	1.1	0.2	1.0	2.1	1.0	2.1	0.0	100%
A3	Financial model, bid strategy, Board/State approvals, bid se...	0.5	1	2	1.1	0.2	2.1	3.2	2.1	3.2	0.0	100%
A4	Technical bid + Initial Offer on MSTC; evaluation; technical...	0.7	1	1.5	1.0	0.1	3.2	4.2	3.2	4.2	0.0	100%
A5	Ascending forward e-auction; preferred bidder; Central Govt...	0.5	1	2	1.1	0.2	4.2	5.3	4.2	5.3	0.0	100%
A6	Execute CMDPA with Nominated Authority	0.3	0.5	1	0.5	0.1	5.3	5.8	5.3	5.8	0.0	100%
A7	1st upfront instalment, performance security, fixed amount...	0.7	1.5	2	1.4	0.2	5.8	7.3	6.7	8.1	0.9	13%
A8	Vesting / allotment order issued (efficiency clock starts)	0.3	0.7	1.5	0.8	0.2	7.3	8.0	8.1	8.9	0.9	13%
B1	Project organisation, PMC, Odisha office, SWCS registration	0.5	1	2	1.1	0.2	5.8	6.9	61.4	62.5	55.5	0%
B2	Tender & onboard consultants (MPPA, NABET EIA, FC/LA/R&R;, DG...	1.5	3	5	3.1	0.6	5.8	8.9	5.8	8.9	0.0	87%
B3	Obtain GR, borehole data & mine dossier (CMPDI / NA)	0.5	1	2	1.1	0.2	8.0	9.1	23.9	25.0	15.9	0%
B4	DGPS boundary survey & CMPDI/ORSAC certification with KML	1	2	4	2.2	0.5	8.9	11.1	8.9	11.1	0.0	100%
B5	Cadastral maps, RoR, plot-wise land schedule certified by RI...	2	3	6	3.3	0.7	11.1	14.4	11.1	14.4	0.0	100%
C1	Mine design, draft Mining Plan + Mine Closure Plan, peer rev...	2	3	5	3.2	0.5	11.1	14.2	25.0	28.2	13.9	0%
C2	SWCS submission; Technical Committee scrutiny & compliance (...	1	1.5	3	1.7	0.3	14.2	15.9	42.5	44.2	28.2	0%
C3	Approval, in-principle approval & final approval of Mining P...	0.5	1	2	1.1	0.2	15.9	17.0	44.2	45.2	28.2	0%
D1	Form-1/CAF + PFR; ToR application on PARIVESH	0.5	1	1.5	1.0	0.2	14.2	15.2	30.5	31.5	16.2	0%
D2	EAC (Coal Mining) appraisal; grant of ToR (60 days)	1	2	4	2.2	0.5	15.2	17.4	33.0	35.2	17.8	0%
D3	Baseline monitoring - one season (air, water, noise, soil, e...	3	3.5	5	3.7	0.3	15.2	18.9	31.5	35.2	16.2	0%
D4	Specialised studies: hydrogeology/GW model, HEC & wildlife...	2	3	5	3.2	0.5	18.9	22.1	35.2	38.3	16.2	0%
D5	Draft EIA/EMP + Odia summary to OSPCB; PH notice (30 days)	1	1.5	3	1.7	0.3	22.1	23.8	38.3	40.0	16.2	0%
D6	Public hearing (within 45 days) and proceedings to MoEFCC	1	1.5	4	1.8	0.5	23.8	25.6	40.0	41.8	16.2	0%
D7	Final EIA/EMP incorporating PH; EC application	1	1.5	3	1.7	0.3	25.6	27.2	41.8	43.5	16.2	0%
D8	EDS/ADS, EAC appraisal & grant of EC (105 days)	2.5	4	10	4.8	1.2	27.2	32.0	43.5	48.2	16.2	0%
F1	Tree enumeration (FMIS maps, committee, DFO approval)	2	3	6	3.3	0.7	14.4	17.8	24.8	28.2	10.4	1%
F2	CA land: identify, Collector NOC, DGPS/DSS, DFO suitability...	3	5	10	5.5	1.2	14.4	19.9	24.3	29.8	9.9	2%
F3	FRA 2006: Gram Sabha resolutions, SDLC/DLC, Collector FRA ce...	3	6	12	6.5	1.5	14.4	20.9	39.8	46.3	25.4	0%
F4	FC proposal (Form-A) on PARIVESH 2.0 with KML, NOCs, draft S...	1	1.5	3	1.7	0.3	17.8	19.4	28.2	29.8	10.4	1%
F5	State processing: DFO/DC, CF, Nodal/PCCF, PSC, State Govt (~...	4	7	14	7.7	1.7	19.9	27.6	29.8	37.5	9.9	3%
F6	IRO inspection, Advisory Committee (FAC), Stage-I in-princip...	2	4	9	4.5	1.2	27.6	32.1	37.5	42.0	9.9	3%
F7	Demand notes & payment (NPV, CA, SMC, WLMP); CA land mutatio...	2	4	8	4.3	1.0	32.1	36.4	42.0	46.3	9.9	3%
F8	Stage-I compliance: schemes, SSWLCP (CWLW), FRA certificate...	3	6	12	6.5	1.5	36.4	42.9	46.3	52.8	9.9	3%
F9	Stage-II (final) approval and State Govt diversion order	1	2	5	2.3	0.7	42.9	45.2	52.8	55.2	9.9	3%
F10	Tree felling (OFDC), working permission & possession of fore...	1	2	4	2.2	0.5	45.2	47.4	55.6	57.7	10.3	1%
L1	Land schedule & requisition to MoC for CBA notifications	1	2	4	2.2	0.5	14.4	16.6	14.4	16.6	0.0	97%
L2	Sec 4(1) notification - intention to prospect	1	3	6	3.2	0.8	16.6	19.8	16.6	19.7	0.0	97%
L3	Sec 7(1) notice of intention to acquire (within 2 yrs of 4(1))	2	4	8	4.3	1.0	19.8	24.1	19.7	24.1	0.0	97%
L4	Sec 8 objections (30 days); hearing & report by Competent Au...	1	2	4	2.2	0.5	24.1	26.2	34.8	37.0	10.7	0%
L5	Sec 9(1) declaration (within 3 yrs of 7(1)); Sec 10 vesting	2	4	8	4.3	1.0	26.2	30.6	37.0	41.3	10.7	0%
L6	Sec 11(1) order vesting land in Govt company; 2nd upfront in...	0.5	1.5	3	1.6	0.4	30.6	32.2	41.3	42.9	10.7	0%
L7	SES, genealogy & YADAST asset enumeration; PAF/PDF lists	6	10	16	10.3	1.7	24.1	34.4	24.1	34.4	0.0	97%
L8	Compensation determination & award (First Schedule via S.O....	2	4	8	4.3	1.0	34.4	38.8	42.9	47.2	8.5	1%
L9	R&R; plan, colony & gochar land; RPDAC approval (ORRP 2006 /...	4	8	15	8.5	1.8	34.4	42.9	34.4	42.9	0.0	96%
L10	Case files, deposit & disbursement of compensation and R&R;	4	8	14	8.3	1.7	42.9	51.2	47.2	55.6	4.3	7%
L11	Government land: alienation/premium payment and handover	3	6	12	6.5	1.5	32.2	38.7	49.1	55.6	16.9	0%
L12	R&R; colony development and shifting of displaced families	8	12	20	12.7	2.0	42.9	55.6	42.9	55.6	0.0	90%
L13	Physical possession of land for rated capacity (first 5 yrs)	1	2	4	2.2	0.5	55.6	57.7	55.6	57.7	0.0	97%
K1	CGWA NOC (NOCAP) for dewatering & GW abstraction	2	3	6	3.3	0.7	22.1	25.4	51.8	55.2	29.7	0%
K2	Water allocation & nala diversion permission (WRD)	3	6	12	6.5	1.5	22.1	28.6	51.2	57.7	29.2	0%
K3	Consent to Establish (OSPCB / OCMMS)	2	3	5	3.2	0.5	32.0	35.2	48.2	51.4	16.2	0%
K4	Power: load sanction, line & substation	4	6	10	6.3	1.0	17.0	23.3	51.4	57.7	34.4	0%
K5	MDO / turnkey contractor: RFP, bidding, evaluation, LoA, con...	6	9	14	9.3	1.3	17.0	26.3	45.2	54.6	28.2	0%
K6	Contractor mobilisation: PBG, site handover, HEMM, workshop	2	3	5	3.2	0.5	35.2	38.3	54.6	57.7	19.4	0%
K7	Approach road & interim road-dispatch set-up (weighbridge, c...	4	6	10	6.3	1.0	35.2	41.5	51.4	57.7	16.2	0%
K8	Explosives (PESO LE-3 magazine) & petroleum storage licences	2	3	5	3.2	0.5	26.3	29.5	54.6	57.7	28.2	0%
K9	DGMS: Form I-A notice (≥30 days), manager appointment, SMP	1	1.5	3	1.7	0.3	26.3	28.0	56.1	57.7	29.8	0%
K10	Consent to Operate (OSPCB)	1.5	2.5	4	2.6	0.4	45.2	47.8	55.2	57.7	9.9	2%
K11	Escrow account for progressive mine closure (CCO beneficiary)	0.5	1	2	1.1	0.2	17.0	18.1	56.7	57.7	39.7	0%
K12	Board approval of mine opening; intimation to CCO; 3rd upfro...	0.5	1	1.5	1.0	0.2	57.7	58.7	57.7	58.7	0.0	100%
K13	Box cut, OB removal and coal exposure	2	3	5	3.2	0.5	58.7	61.9	58.7	61.9	0.0	97%
K14	Commercial readiness: royalty/DMF/NMET, i3MS e-permit, grade...	1	2	3	2.0	0.3	58.7	60.7	59.9	61.9	1.2	3%
K15	First coal production & dispatch; 4th upfront instalment	0.3	0.5	1	0.5	0.1	61.9	62.5	61.9	62.5	0.0	100%

Time-scaled activity-on-node network



Each node is placed at its earliest start (months from T0) in its phase lane; the label gives t_e. Red nodes and links form the critical path; navy diamonds mark CMDPA milestones.

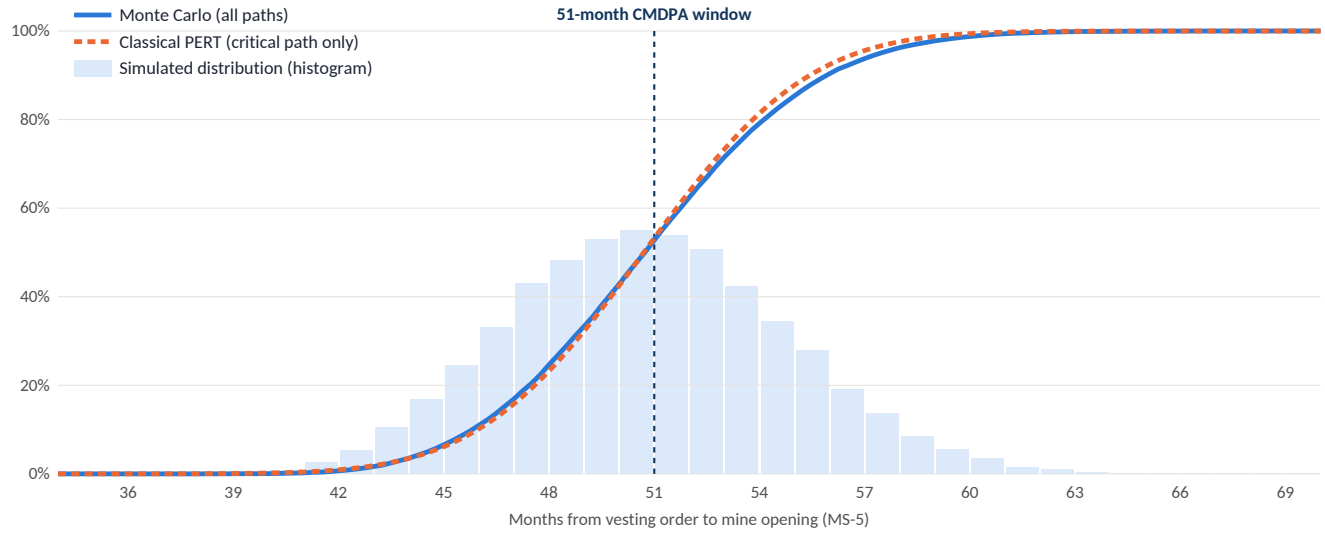
Gantt chart with float



Bars: ES to EF (t_e); dotted tail: total float up to LF; red: critical; diamonds: milestones.

RISK-ADJUSTED SCHEDULE

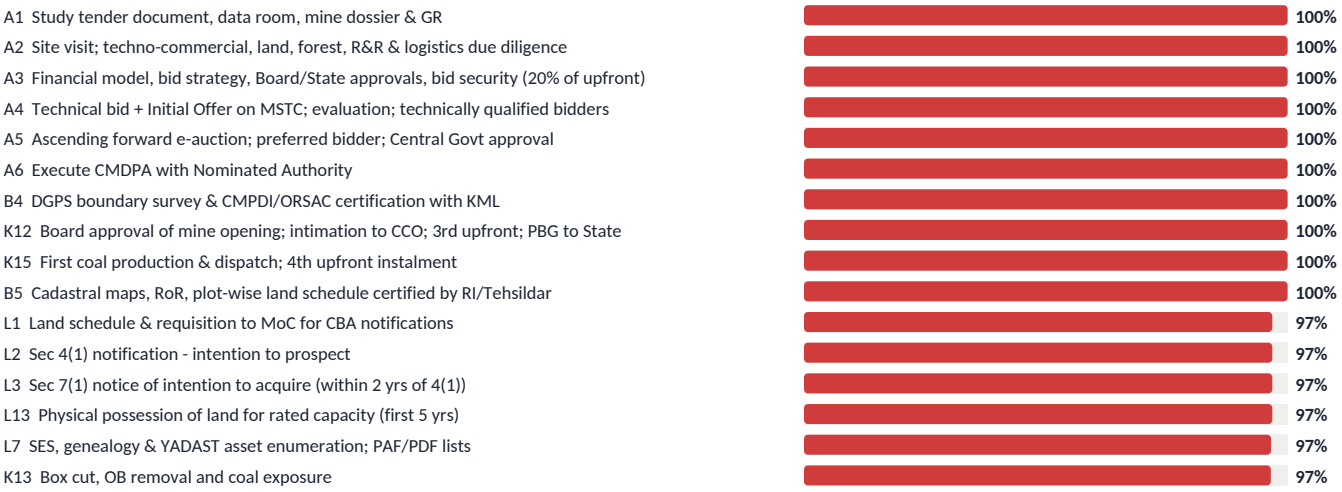
8 Probability, criticality and acceleration



Cumulative probability of reaching MS-5 by a given month after vesting. The PERT curve ignores the near-critical forest path and is therefore slightly optimistic (merge bias).

Target (months after vesting)	42	45	48	51	54	57	60
Monte Carlo probability	1%	7%	25%	53%	79%	94%	99%
PERT normal approximation	1%	6%	23%	53%	81%	96%	99%

Criticality index - which activities really drive the finish



Share of 20,000 simulated schedules in which the activity lies on the longest path.

Acceleration scenarios (crashing and fast-tracking)

	Scenario	Expected	P50	P80	P(≤51 m)
S0	Baseline - CMDPA sequence (mine opening after full MS-4 possession incl. R&R; shifting)	50.7	50.7	54.2	53%
S1	Early SES: start SES/genealogy at s.4(1) instead of s.7(1)	46.4	46.7	49.9	87%
S2	R&R; plan & RPDAC prepared in parallel with SES (overlap ~50%)	46.7	47.0	50.0	87%
S3	Phased possession: box cut on a no-habitation patch; colony & shifting continue	44.2	44.9	47.8	95%
S4	S1 + S2 + S3 combined	40.8	41.8	44.5	100%
S5	S4 + CA land pre-identified at bid stage + dedicated FC liaison	37.8	39.0	41.3	100%

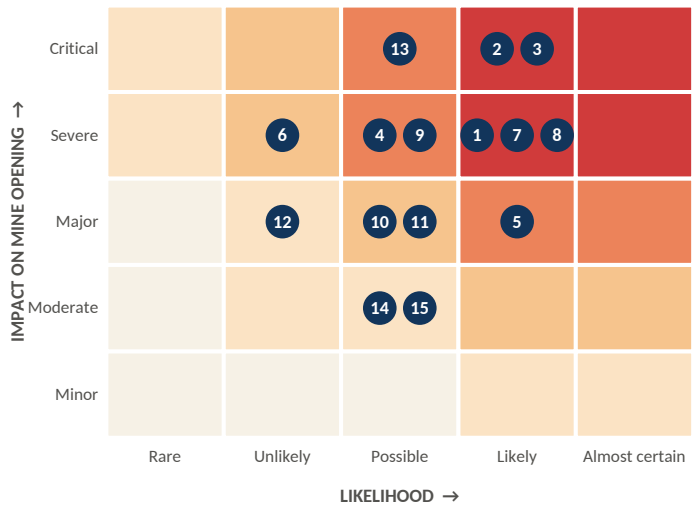
Months from vesting to MS-5, 12,000 runs per scenario. Once land is accelerated (S3-S5), the forest-clearance chain (CA land → State processing → Stage-I compliance → Stage-II) becomes critical - so land and forest fast-tracking must be done together.

WHAT TO WATCH

9 Critical activities register

#	Activity (ID)	Why it is critical	Early-warning sign	Mitigation / owner
1	SES, genealogy, YADAST (L7)	Defines PAF/PDF base; disputes stall RPDAC (BW: recorded-tenant basis)	Families surveyed vs estimate; objections logged	Start at s.4(1); drone + GIS; supplementary survey of actual families - LA agency / Collector
2	R&R plan & RPDAC (L9)	RPDAC dates depend on the Collector; colony and gochar land must be fixed	No RPDAC date 60 days after SES	Pre-identify colony land at bid stage; RDC escalation; draft minutes ready - GMDC
3	R&R colony & shifting (L12)	Longest single activity (8-20 months)	Colony civil progress < plan	Phase colony; cash-in-lieu options; shift initial patch first - GMDC / MDO
4	CBA s.4(1) & s.7(1) (L2, L3)	Gazette queue at MoC; schedule errors restart the clock	Draft notification returned	Tehsildar-certified schedule; CBA-cell liaison; parcel-wise s.9 - GMDC
5	Compensation disbursement (L10)	Title disputes, mutation backlog, monthly rate revisions	Case files validated vs target	Village-wise camps; deposit disputed sums with Tribunal (s.17) - Collector
6	DGPS & land schedule (B4, B5)	Every clearance consumes the certified boundary and schedule	Certificate not in hand at vesting + 3 m	Order DGPS on day 1 after CMDPA; ORSAC - PMC
7	CA land (F2)	Suitability, mutation, PF notification; often in another district	No suitability certificate by month 6	Search at bid stage; accredited CA land bank - GMDC / DFO
8	State FC processing (F5)	Five desks; EDS loops; ~100 working days at best	File idle >15 working days at a desk	Desk-wise liaison; PARIVESH tracking; complete checklist - FC consultant
9	Stage-I compliance (F8)	12 schemes, SSWLCP, FRA certificate, extra CA cost	Schemes approved vs total	Prepare schemes in parallel with Stage-I; pre-fund demands - GMDC
10	FRA certificate (F3)	Prerequisite for State diversion order	Gram Sabha quorum failures	Early Gram Sabhas, video minutes, SDLC/DLC calendar - Collector
11	EC appraisal (D8)	EAC may return/defer (BW: 03-Apr-25)	EDS count >1; adverse PH issues	Mock EAC; elephant-corridor, cumulative-impact and R&R chapters upfront - EIA consultant
12	CTO (K10)	Waits for FC Stage-II; gates mine opening	CTO queries open after FC-II	Apply early; close all queries except FC-II - GMDC
13	Mine-opening gate (K12)	Ten inputs converge; any one blocks MS-5	Checklist items open at M-3	Readiness board from M-6; Board paper template - PMC
14	Nala diversion / water (K2)	WRD field verification and NOC	No joint visit 60 days after DPR	Early DPR; Chief Engineer escalation - GMDC
15	Rail siding (R1-R3)	Needed before 6 MTPA; 24-36 months	DPR not approved 30 months before 6 MTPA year	Parallel land for siding; ECoR follow-up - GMDC / IPRCL

Risk heat map (impact on mine-opening date)



Numbers refer to the register above.

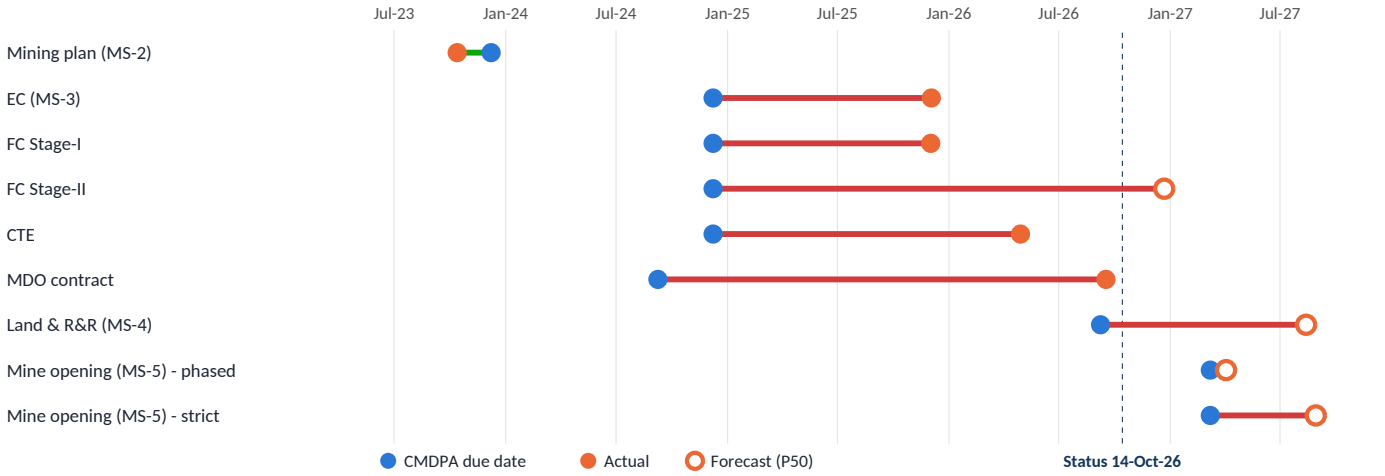
Top-right cells (likely and severe) are the R&R chain (2, 3), the land base (1), CA land and State FC processing (7, 8) and the mine-opening gate (13). These deserve weekly PMC review, named owners at GM level and escalation triggers.

Lower-left items still need tracking but can be managed monthly.

APPLYING THE MODEL

10 Baitarani West - status and forecast

Status reconstructed from the MIS of 30-Sep-2026 and later updates (FC Stage-I compliance file forwarded by PCCF to MoEFCC on 08-Oct-26; RPDAC held 07-Oct-26). Months are offsets from the vesting order (08-Jun-2023).



Blue dot = due date (CMDPA basis; MDO from the MD's internal plan); orange = actual (solid) or P50 forecast (hollow); red bar = delay. FC Stage-II forecast uses the expected critical chain.

Remaining-work network (from 14-Oct-2026)

ID	Remaining activity	Owner	a	m	b	te	EF date	TF	CI
W1	MoEFCC processing of Stage-I compliance; Stage-II approval	MoEFCC	0.7	2	5	2.3	22-Dec-26	0.0	52%
W2	FRA certificate / NOC from DLC & Collector (overdue)	Collector Angul	0.5	1.5	4	1.8	06-Dec-26	0.5	22%
W3	Nala diversion NOC from WRD	WRD Odisha	0.3	1	3	1.2	20-Nov-26	3.4	0%
W4	State Govt diversion order (after Stage-II & FRA)	Forest Dept	0.3	0.7	1.5	0.8	14-Jan-27	0.0	74%
W5	Tree felling & possession of 64.37 ha forest land	DFO / OFDC	0.7	1.5	3	1.6	05-Mar-27	0.0	59%
W6	R&R; scheme finalisation after RPDAC (07-Oct-26)	Collector / RDC	0.5	1.5	3	1.6	01-Dec-26	0.4	1%
W7	Case files (199 ha) & compensation disbursement, 5-yr area	BNB / Collector	1	2.5	6	2.8	08-Jan-27	0.6	26%
W8	Physical possession of 5-yr private & Govt land	Collector / GMDC	0.5	1	3	1.2	15-Feb-27	0.6	26%
W9	R&R; colony land, development & shifting of PDFs	GMDC / Collector	4	8	15	8.5	16-Aug-27	0.4	0%
W10	Consent to Operate (pending FC Stage-II)	OSPCB	0.5	1	2.5	1.2	19-Feb-27	0.4	15%
W11	MDO mobilisation (HEMM, workshop, magazine)	MDO	0.5	1.5	3	1.6	01-Dec-26	3.1	0%
W12	DGMS notices & appointments; explosives/HSD licences	DGMS / PESO	0.5	1.5	3	1.6	01-Dec-26	3.1	0%
W13	Tree cutting on Govt & acquired private land	Forest / GMDC	0.5	1	2.5	1.2	18-Nov-26	3.5	0%
W14	Board approval of mine opening; CCO intimation; 3rd upfront	GMDC Board	0.2	0.5	1	0.5	21-Mar-27	0.0	100%
W15	Box cut & start of OB removal	MDO	0.5	1	2	1.1	23-Apr-27	0.0	100%
W16	Coal exposure & first coal dispatch by road	GMDC / MDO	2	4	7	4.2	28-Aug-27	0.0	100%

Critical chain (phased possession): W1 → W4 → W5 → W14 → W15 → W16. W9 (R&R colony and shifting) is off the path only if the box cut is placed on a patch with no habitation.

Forecast (Monte Carlo, 20,000 runs)	Phased possession (box cut on cleared patch)	Strict sequence (R&R shifting before MS-5)
MS-5 mine opening - P50 / P80	03-Apr-27 / 24-Apr-27	29-Aug-27 / 29-Oct-27
P(MS-5 by CMDPA date 08-Mar-27)	12%	0%
P(MS-5 within 51-month window, 08-Sep-27)	100%	55%
First coal - P50 / P80	10-Sep-27 / 13-Oct-27	05-Feb-28 / 10-Apr-28

Two checks the MIS does not show

Rail check. With IPA in hand (18-Jan-25) the remaining DPR/ESP approvals and construction take t_e ≈ 29 months → siding ready ≈ **Mar-2029**. Production reaches 6 MTPA about 24 months after first coal → needed by ≈ **Sep-2029**. Float is only ~6 months: rail is near-critical for ramp-up and for the early-production rebate.

MIS forecast vs model. The MIS high-level chart forecasts mine opening around 15-Nov-26; the model suggests this needs FC Stage-II, the State order, CTO and possession of the box-cut land to close within ~5 weeks - possible only if every item hits its optimistic time.

BEYOND THE BRIEF

11 Commercial levers you may not be using

Lever	What the CMDPA / rules allow	So what for BW
Refund of appropriated PS	Schedule D note 6: if MS-5 is achieved within 51 months, PS appropriated for earlier delays may be refunded	₹386.8 Cr (MS-3 10% + MS-4 25%) recoverable if MS-5 by 08-Sep-27 - the strongest reason to protect the mine-opening date
Extension for bona fide delays	Schedule D note 4: NA may extend timelines via the Scrutiny Committee for delays not attributable to the bidder	Document State-side delays (2nd upfront after s.11(1), EAC return, CA norm change, FRA) and file a reasoned representation
Early-production rebate	Cl. 9.2: 50% rebate on Final Offer for coal produced before the Scheduled Date of Production (75 months from vesting)	Window closes 08-Sep-2029. Each Mt saves ₹16.75 Cr for every ₹1,000/t of notional price; front-load years 3-5
Gasification incentive	Cl. 9.3: 50% rebate on coal used/sold for gasification if ≥10% of scheduled production	Explore a gasification/derivatives tie-up for G12 coal (not combinable with cl. 9.2 on the same tonnes)
Minor mining-plan changes	MP Guidelines 2025 para 2.9b: Board can approve ≤50% capacity increase, HEMM or infrastructure relocation	Re-site box cut/infrastructure away from habitation; raise early-year output without full revision
Production flexibility	Cl. 16.2: ≥65% of scheduled output in a year and ≥75% over 3-year blocks	Plan ramp-up and rail to stay above 65% from the first scheduled year
Upfront adjustment	Cl. 9.4: upfront adjusted against monthly payments, capped at 50% per month	₹500 Cr is recovered from revenue share - include in the cash-flow model
PBG revision timing	Cl. 6.1.4: first revision at mine opening on NCI; then every April	Coordinate BG limits with banks before MS-5
Colliery Control amendment	Board approves mine opening; intimate CCO within 15 days	Saves up to ~2 months vs the old CCO permission; prepare the Board note in advance

USING THE OLD TRACKER PROPERLY

12 Reconciliation of 'Final Tracker_cl.xlsx'

The workbook (sheets BP, BW_, Partner Onboarding, Support Activities, HO Core Team) dates from mid-2023. Its activity lists are useful; its dates are stale and several procedures have changed. Item-by-item:

Tracker block	What it says	Current position / correction
A1 Contract formalisation	'Agreement signing 08-Jun-23'	CMDPA signed 29-Mar-23; 08-Jun-23 is the vesting order date
P1-P6 Preliminary	DGPS Aug-Oct-23; GR soft copy 13-Jul-23	DGPS by CMPDIL completed 12-Jan-24; continue to use for CA land and siding surveys
MP1-MP14 Mining plan	Fresh plan, MoC approval Nov/Dec-23	Not needed: OMC plan (10-Nov-21) deemed transferred; MS-2 met. Revise only for optimisation
FC1-FC30 Forest clearance	Old Part I-V file movement (FCR 2003)	Superseded by VSSR 2023 / PARIVESH 2.0; Stage-I 02-Dec-25; Stage-II pending at MoEFCC
FRA1-FRA9	Collector-led FRA process	Still valid; certificate pending - now on the critical path for the State order
CA1-CA13 / TE1-TE10 / WLCP1-4	CA land, tree enumeration, wildlife plan	Done: TE (Jun-24), CA land Koraput mutated & PF-notified (11-Jun-26), SSWLCP and elephant-corridor NOC (25-Aug-25)
EC1-EC19	Form-1 Oct-Nov-23 onwards	ToR transferred 06-Nov-23; EC granted 03-Dec-25
ML1-ML12 Mining lease	State ML via Collector, DGM, MRD, MoC prior approval	Not required for CBA land: s.11(1) order (13-Oct-23) substitutes for ML; confirm whether State wants a lease for Govt/forest parcels
LA1-LA50 Land & R&R	RFCTLARR 2013 s.4 SIA, s.11, s.19, s.21, s.23, s.38	Wrong statute for this route - replace with CBA s.4(1)/7(1)/8/9(1)/10/11(1)/12/13-14/17 plus RFCTLARR Sch. I-III (S.O. 2368(E)) and ORRP 2006 RPDAC
CTE1-CTE6 / CTO1-CTO5	~60 days each	CTE took ~4 months (02-Jan to 29-Apr-26); CTO applied 08-Jul-26, pending FC-II
MO1-MO9 Mine opening	'Grant of mine opening permission by Coal Controller'	Replaced by Board approval + CCO intimation (Colliery Control (Amendment) Rules 2025/2026); escrow done
Partner Onboarding	Tenders for MP, EC+FC, LA, DPR, MDO (2023)	All agencies onboard; MDO LoI 13-Nov-25, contract 17-Sep-26
HO Core Team deadlines	EC/FC 07-Dec-24; LA 07-Sep-26; opening 07-Mar-27	EC met 12 months late; MS-4 overdue; MS-5 at risk - see Section 10
Missing altogether	-	CGWA NOC, WRD water & nala NOC, power, PESO licences, DGMS notices & manager, CMPF/CLRA, royalty/DMF/NMET/i3MS, grade declaration, GCT rail approvals, approach road, RPDAC, SES/genealogy/YADAST, R&R colony & gochar land, upfront 3rd/4th, PBG transfer, extension request

WHAT TO DO NEXT

13 Recommendations and 90-day action plan

#	Action	Owner	By
1	Daily follow-up of FC Stage-I compliance at MoEFCC (FC division) and IRO; pre-empt queries	GMDC FC cell / DCS	Nov-26
2	Obtain FRA certificate (DLC Angul) - prerequisite for the State diversion order	GMDC / Collector	Oct-26
3	Secure nala-diversion NOC from WRD (joint report submitted 29-Apr-26)	GMDC	Oct-26
4	Close all CTO observations except FC-II so CTO issues within days of Stage-II	GMDC Env cell	Nov-26
5	Finalise and validate 199-ha case files; village-wise disbursement camps; execute Govt-land possession	BNB / Collector	Dec-26
6	Notify R&R scheme after RPDAC; lock R&R colony and gochar land; start colony civil works	Collector / GMDC	Jan-27
7	Supplementary SES for extended/actual families to de-risk R&R entitlements	BNB	Dec-26
8	Fix the box-cut patch on Govt/acquired land with no habitation; complete tree cutting there	GMDC / MDO	Nov-26
9	DGMS Form I-A notice, manager appointment, SMP; PESO magazine & HSD licences	MDO / GMDC	Dec-26
10	Mine-opening readiness board (all approvals) and Board note; fund 3rd upfront ₹125 Cr and PBG transfer	PMC / Finance	Jan-27
11	File extension representation to NA for delays not attributable to GMDC	GMDC / PMC	Dec-26
12	Push ECoR DPR/ESP approval; acquire siding and coal-stock land at Jharpada	GMDC / IPRCL	Mar-27
13	Customer contracts and grade declaration; royalty/DMF/NMET and i3MS registrations	Marketing / Finance	Feb-27
14	Enforce contractual LDs on agencies that missed milestones, without stalling deliverables	GMDC Contracts	Ongoing
15	Replace the 2023 tracker with the activity list in Section 5 and this report's network	PMC	Nov-26

Governance rhythm the PMC should run

- **Weekly**: critical-chain stand-up (land, R&R, FC-II, CTO, possession) with named owners and a two-week look-ahead.
- **Fortnightly**: desk-level tracking of every file in Government (date received, days idle, next action) - VSSR and the Mining Plan Guidelines publish desk timelines that can be cited in escalations.
- **Monthly**: re-run this PERT/CPM model with actuals; publish P50/P80 dates for MS-4, MS-5 and first coal; report the efficiency-parameter status to the Nominated Authority.
- **Quarterly**: Board-level review of performance-security exposure, the 51-month window and the early-production rebate.

Lessons for the next block

For a **new** Odisha block, the biggest gains come before and immediately after the bid: price and plan R&R and CA land into the bid, order DGPS and the land schedule on day 1, run land and forest tracks in parallel from s.4(1), and design the first-year box cut away from habitation.

ANNEX

A Glossary

Term	Meaning	Term	Meaning
CBA	Coal Bearing Areas (Acquisition & Development) Act, 1957	MS-2 ... MS-5	CMDPA efficiency-parameter milestones
CCO	Coal Controller's Organisation (MoC)	NA	Nominated Authority, Ministry of Coal
CMDPA	Coal Mine Development & Production Agreement	NCI	National Coal Index
CA / NPV	Compensatory afforestation / Net Present Value of forest	PAF / PDF	Project affected / displaced family
CTE / CTO	Consent to Establish / Operate (OSPCB)	PBG / PS	Performance bank guarantee / performance security
CPM / PERT	Critical Path Method / Programme Evaluation & Review Technique	PRC	Peak rated capacity
DLC / SDLC	District / Sub-Divisional Level Committee (FRA)	RPDAC	Rehabilitation & Periphery Development Advisory Committee
EAC / FAC	Expert Appraisal Committee / Forest Advisory Committee	SES	Socio-economic survey
EDS / ADS	Essential / Additional details sought	SSWLCP	Site-specific wildlife conservation plan
ES, EF, LS, LF	Earliest/latest start and finish of an activity	SWCS	Single Window Clearance System (MoC)
GCT / ESP / IPA	Gati Shakti Cargo Terminal / Engineering Scale Plan / In-principle approval	TF / FF	Total float / free float
IRO	Integrated Regional Office of MoEFCC	VSSR	Van (Sanrakshan Evam Samvardhan) Rules, 2023
MDO / HEMM	Mine Developer & Operator / Heavy Earth-Moving Machinery	YADAST	Odisha enumeration of houses, structures & assets for valuation
MPPA / QP	Mining Plan Preparing Agency / Qualified Person		

B Sources

Web sources (retrieved October 2026 via Firecrawl)

#	Source	Location
1	PIB (26-Dec-2025) - Colliery Control (Amendment) Rules, 2025: Board of company to approve mine opening	pib.gov.in/PressReleasePage.aspx?PRID=2208844
2	Corpseed note - Colliery Control (Amendment) Rules, 2026 (intimation to CCO within 15 days)	corpseed.com/news/colliery-control-amendment-rules-2026-coal-mine-approval
3	Coal Controller's Organisation - Opening & re-opening of coal mines (documents, SWCS process)	coalcontroller.gov.in/opening-and-re-opening-coal-mines-and-coal-seams
4	Ministry of Coal - Single Window Clearance System	coal.nic.in/nominated-authority/single-window-system
5	MoC - Mining Plan Guidelines for Coal & Lignite Mines 2025 (Jan-2025)	coal.nic.in/sites/default/files/2025-01/31-01-2025a-wn.pdf
6	Van (Sanrakshan Evam Samvardhan) Rules, 2023 - timelines schedule	thc.nic.in/Central Governmental Rules/Van (Sanrakshan Evam Samvardhan) Rules, 2023.pdf
7	MoC - Minutes of NA review of coal-block operationalisation (Baitarni West, Burapahar et al.)	coal.gov.in/sites/default/files/2023-06/24-07-2025mom.pdf
8	Madras HC, G. Kesavan v. UoI (2025) - S.O. 2368(E) applies RFCTLARR Schedules to CBA	casemine.com/judgement/in/67c14327ed57983ce54e4142
9	Railway Board - GCT Master Circular 2026	indianrailways.gov.in/railwayboard/.../GCT Master Circular 2026.pdf
10	PIB (17-Sep-2026) - 16th round of commercial coal mine auctions; 147 mines auctioned	pib.gov.in/PressReleaseIframePage.aspx?PRID=2311453
11	GMDC Annual Report 2025-26 (NSE filing) - Baitarni West status	nsearchives.nseindia.com/annual_reports/AR_31488_GMDCLTD_2025_2026
12	Construction World (12-Dec-2025) - GMDC secures EC and FC Stage-I; mining partner onboarded	constructionworld.in/.../gmdc-advances-rs-15-mtpa-baitarni-west-mine
13	CGWA - documents required for mining projects	cgwa.mowr.gov.in/DocRecMining.html
14	Odisha SPCB - consent to operate	ospcboard.odisha.gov.in/en/divisions/consent-management-division
15	Odisha R&R; Policy 2006 (Revenue & DM Dept)	revenue.odisha.gov.in/sites/default/files/2020-05/18040_14_05_06.pdf

Project documents in the repository

- CMDPA for Baitarni West (29-Mar-2023) - Final Offer, upfront, PBG, Schedule D efficiency parameters
- Final EIA/EMP, Baitarni West OCP 15 MTPA (GRC India) - project facts, land, reserves, production plan
- MIS Report of Burapahar & Baitarani West, 30-Sep-2026 (PMC) - actual and forecast status
- Final Tracker_cl.xlsx - 2023 activity lists (reconciled in Section 12)
- Bidding presentation (MoC, 10th tranche) - auction mechanics, payments and guarantees
- CBA Act 1957; CM(SP) Act 2015; MMDR Act 1957; EIA Notification 2006; EP Act 1986
- RFCTLARR Act 2013; ORFCTLARR Rules 2016; Odisha R&R; Policy 2006; FRA 2006 and Rules; PESA 1996; Companies Acts

Statutory limits are as published at retrieval; three-point durations are professional estimates to be refreshed monthly with actuals. Not legal advice.